

2024, Mar. 2024 Release LOC:Acute Adult Extended Stay

Overview

Select Level of Care

Acute ⁽¹⁾

Intermediate ⁽¹²²⁾

Critical ⁽¹⁴³⁾

Discharge Screens ⁽¹⁸²⁾

Notes

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Overview

Instruction: This subset is for patients who are not clinically stable for discharge, but exhaust the episode days for their primary condition within a condition-specific or general subset. Patients who are treated in this subset should receive discharge planning and demonstrate progress towards transfer to a post acute level of care.

If a patient develops a new condition necessitating continued stay, the reviewer should refer to Episode Day 1 or Operative Day within the most appropriate subset. For example:

- If a patient develops pneumonia during the extended hospitalization, refer to the Infection: Pneumonia subset.
- If a patient requires surgical intervention during the extended hospitalization, refer to the General Surgical subset.

If a patient does not meet criteria in any appropriate subset, refer for secondary review.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Level of Care, One:● **ACUTE, One:** ⁽¹⁾● **General, One:**● **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽²⁾● **Clinical stability, ≥ One:**

- Anemia controlled or resolved
- Bleeding controlled or resolved
- Blood glucose within acceptable limits ⁽³⁾
- Hct or Hb within acceptable limits ⁽³⁾

● **Functional impairment and, One:** ⁽⁴⁾● **Discharge to home and, One:** ⁽⁵⁾

- Patient or caregiver able to safely manage impairment
- Home care services arranged
- Discharge to inpatient facility for rehabilitative services
- Pain controlled or manageable ⁽⁶⁾

● **Therapeutic anticoagulation, ≥ One:**

- Anticoagulation regimen established and laboratory values within acceptable limits ⁽⁷⁾
- IVC filter

● **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**● **Bleeding and, Both:**● **Finding, ≥ One:**

- Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) ⁽⁸⁾
- Exertional dyspnea worse than baseline ^(9, 10)
- Hct or Hb decreased last 24h
- Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽¹¹⁾

● **Orthostatic hypotension, Both:** ⁽¹²⁾

- Sustained drop in blood pressure within 3 min of sitting or standing
- Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
- Syncope ⁽¹³⁾

● **Requiring Hct or Hb monitoring at least 2x/24h, ≤ 2d, ≥ One:**

- Blood product transfusion ⁽⁸⁾
- Return to OR within last 24h

● **Blood glucose ≤ 24h, All:** ⁽¹⁴⁾● **Finding, ≥ One:**

- > 250 mg/dL(13.9 mmol/L)
- < 70 mg/dL(3.9 mmol/L)
- Blood glucose monitoring
- Insulin (includes SC) adjustment or dose held ^(14, 15)

● **Functional impairment (new) requiring rehabilitation initiation ≤ 24h and, ≥ One:** ^(4, 16)

- PT evaluation and training
- OT evaluation and training
- Speech-language evaluation and training
- Cognitive evaluation and retraining
- Swallowing evaluation and retraining

● **Hemolytic anemia and, Both:** ⁽¹⁷⁾

- Hct < 30%(0.30) or Hb < 10.0 g/dL(100 g/L)

● **Hct or Hb monitoring at least 2x/24h and, ≥ One:**

- Blood product transfusion ⁽⁸⁾
- Corticosteroid ⁽¹⁸⁾
- Immunotherapy (includes monoclonal antibody treatment) ≤ 24h
- Immunoglobulin ⁽¹⁹⁾
- Plasmapheresis

- HIT and anticoagulation ^(20, 21)
- Pain uncontrolled and, ≥ **One:** ^(22, 23)
 - Analgesic, continuous
 - Analgesic ≥ 3x/24h
 - Requiring change in medication, dose, or frequency ≤ 2d ⁽²⁴⁾
 - Transition to PO analgesic ≤ 24h
- Pretreatment or bridging anticoagulation, ≥ **One:** ⁽²⁵⁾
 - Compliance risk, **All:**
 - Patient or caregiver unable to administer
 - Outpatient services unavailable ⁽²⁶⁾
 - INR non-therapeutic and anticoagulation (includes PO) ⁽²⁷⁾
 - UFH, continuous and INR non-therapeutic
 - Therapeutic anticoagulation ≤ 5d since initiation ^(28, 29)
 - Post Critical care ≤ 24h
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
- **Cardiovascular or peripheral vascular, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽²⁾
 - Clinical stability, ≥ **One:**
 - Heart failure resolving or returned to baseline ⁽¹⁰⁾
 - Pericardial effusion resolving and pain controlled or manageable
 - Medication regimen established and tolerated
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Acute heart failure unresolved ≤ 3d and, **Both:** ⁽³⁰⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ^(30, 31)
 - Intervention, ≥ **One:**
 - Diuretic, **One:** ⁽³²⁾
 - ≥ 2x/24h (includes PO)
 - Requires daily medication adjustment (includes PO) and BUN or creatinine increased from baseline ^(10, 24)
 - Dialysis (includes ultrafiltration)
 - Inotrope ≤ 4d since initiation and, **Both:** ^(33, 34, 35)
 - Continuous infusion
 - Requiring infrequent titration
 - Pericardial drain removed ≤ 24h
 - PO medication adjustment ≤ 2d and, ≥ **One:** ⁽²⁴⁾
 - ACE inhibitor
 - Antiarrhythmic
 - Antihypertensive
 - Calcium channel or beta blocker
 - Vasodilator or inotrope ⁽³⁶⁾
 - Post Critical care ≤ 24h
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
- **Electrolyte or mineral imbalance, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽²⁾
 - Electrolytes within acceptable limits
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Hypercalcemia, **Both:** ^(37, 38)
 - Calcium 11.1-13.9 mg/dL(2.78-3.48 mmol/L)
 - Intervention, ≥ **One:**
 - Bisphosphonate ⁽³⁹⁾
 - Corticosteroid (includes PO) ⁽³⁹⁾

- Saline hydration ^(40, 41)
- **Hypocalcemia, Both:** ^(38, 42)
 - Calcium 5.0-7.4 mg/dL(1.25-1.85 mmol/L)
 - Calcium repletion
- **Hyperkalemia, All:**
 - Potassium > 5.5 mEq/L(5.5 mmol/L)
 - No electrocardiogram (ECG) changes ^(43, 44)
- **Intervention, ≥ One:**
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose 50%(0.50) with insulin ⁽⁴⁵⁾
 - IV fluid ⁽⁴¹⁾
 - Potassium binding agent (includes PO or PR) ⁽⁴⁶⁾
- **Hypokalemia, All:** ^(47, 48)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - No electrocardiogram (ECG) changes ^(43, 44)
 - Potassium repletion (includes PO)
- **Hypermagnesemia, Both:** ⁽⁴⁹⁾
 - Magnesium > 2.5 mg/dL(1.03 mmol/L)
- **Intervention, ≥ One:**
 - Diuretic ≥ 2x/24h (includes PO)
 - IV fluid ⁽⁴¹⁾
- **Hypomagnesemia, Both:**
 - Magnesium 1.0-1.4 mg/dL(0.41-0.58 mmol/L)
 - Magnesium repletion
- **Hypernatremia, Both:** ⁽⁵⁰⁾
 - Sodium 151-160 mEq/L(151-160 mmol/L)
 - IV fluid ⁽⁴¹⁾
- **Hyponatremia or SIADH, Both:** ⁽⁵¹⁾
 - Sodium 120-129 mEq/L(120-129 mmol/L)
- **Intervention, ≥ One:**
 - Diuretic and oral sodium supplement
 - Fluid restriction
- **Medication administration, ≥ One:**
 - Demeclocycline (includes PO)
 - Diuretic ≥ 2x/24h
 - Sodium chloride 0.9%(0.009) ⁽⁵²⁾
- **Hypophosphatemia, Both:** ⁽⁵³⁾
 - Phosphate 1.0-2.0 mg/dL(0.32-0.65 mmol/L)
 - Phosphate repletion (includes PO)
- Post Critical care ≤ 24h
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
- **Gastrointestinal or biliary, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽²⁾
 - **Clinical stability, ≥ One:**
 - Ascites controlled or improving and tolerating PO or nutritional route established ^(54, 55)
 - **Bilirubin and, All:**
 - Laboratory values normal or within acceptable limits ⁽³⁾
 - Neurological stability ⁽⁵⁶⁾
 - Tolerating PO or nutritional route established ^(54, 55)
 - **Hepatic encephalopathy, Both:** ⁽⁵⁷⁾
 - Grade I or within acceptable limits ⁽³⁾

- Neurological stability last 24h ⁽⁵⁶⁾
- Ileus resolved and tolerating PO or nutritional route established ^(54, 55)
- Tolerating PO or nutritional route established ^(54, 55)
- Upper gastrointestinal (GI) non-variceal bleeding, **Both**:
 - Bleeding controlled or resolved
 - High-dose proton pump inhibitor regimen completed or established
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Ascites, ≤ 7d since initiation of treatment and, **Both**:
 - Finding, ≥ **One**:
 - Abdominal girth or weight, ≥ **One**:
 - Increasing
 - Subtherapeutic response to treatment ⁽⁵⁸⁾
 - Unchanged from admission
 - Dyspnea > baseline ⁽¹⁰⁾
 - Intervention, ≥ **One**:
 - Albumin
 - Diuretic ≥ 2x/24h
 - Sodium and fluid restriction ≤ 3d
 - Therapeutic paracentesis performed within last 24h
 - Transjugular intrahepatic portosystemic shunt (TIPS) performed ≤ 24h
 - Bilirubin > 3.0 mg/dL(51.3 μmol/L) and, ≥ **One**:
 - Albumin level < 2.0 g/dL(20 g/L) and, ≥ **One**:
 - Albumin
 - Feeding tube insertion performed within last 24h and feedings initiated
 - TPN or PPN
 - Inadequate oral intake ≤ 4d or active vomiting and, ≥ **One**: ⁽⁵⁹⁾
 - Antiemetic ≥ 3x/24h ⁽⁶⁰⁾
 - Serotonin antagonist > 2x/24h ⁽⁶¹⁾
 - IV fluid ⁽⁴¹⁾
 - Liver biopsy results pending ≤ 24h
 - PT ≥ 1.5x ULN or INR ≥ 2.0 and fresh frozen plasma (FFP)
 - Hepatic encephalopathy Grade II ≤ 7d since initiation of treatment and, ≥ **One**: ^(57, 62, 63)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽⁶⁴⁾
 - Lactulose (includes PO or PR) ⁽⁶⁵⁾
 - L-ornithine L-aspartate (LOLA) (includes PO) ⁽⁶⁶⁾
 - Neomycin (includes PO) ⁽⁶⁷⁾
 - Rifaximin (includes PO) ⁽⁶⁸⁾
 - Ileus, **One**: ⁽⁶⁹⁾
 - Resolved and initiating PO or tube feeding ≤ 2d
 - Unresolved and ≤ 5d since initiation of treatment and, ≥ **One**: ⁽⁷⁰⁾
 - IV fluid ⁽⁴¹⁾
 - TPN or PPN
 - Inadequate oral intake and, ≥ **One**: ⁽⁵⁹⁾
 - IV fluid ≤ 2d ⁽⁴¹⁾
 - Parenteral nutrition ≤ 2d
 - Initiation or adjustment of PO or tube feeding ≤ 2d
 - Post Critical care ≤ 24h
 - Upper gastrointestinal (GI) non-variceal bleeding and high-dose proton pump inhibitor ≤ 3d since initiation ⁽⁷¹⁾
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
- **Genitourinary, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**: ⁽²⁾

- **AKI, $\geq$ One:**
 - Kidney function improving or within acceptable limits ⁽³⁾
 - Outpatient dialysis regimen established
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - **AKI, Both:** ⁽⁷²⁾
 - **Finding, $\geq$ One:**
 - Oliguria ^(73, 74)
 - **Creatinine, One:**
 - $\geq 1.5\times$ baseline and chronic kidney disease
 - $\geq 1.5\times$ ULN ⁽⁷⁵⁾
 - GFR $> 25\%(0.25)$ decrease from baseline ⁽¹⁰⁾
 - Dialysis initiated
 - **Intervention, $\geq$ One:**
 - Volume expander $\leq 2d$ ^(76, 77)
 - Diuretic $\geq 2x/24h$, $\leq 3d$
 - Dialysis (initial) $\leq 7d$ since initiation ^(78, 79)
 - Dialysis discontinued requiring lab monitoring $\leq 2d$ since discontinuation ⁽⁸⁰⁾
 - Immunosuppressant therapy $\leq 3d$ ^(81, 82, 83)
 - Post Critical care $\leq 24h$
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
- **Infection, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁾
 - Afebrile
 - Laboratory values within acceptable limits ⁽³⁾
 - Tolerating PO ⁽⁵⁴⁾
 - **Clinical stability, $\geq$ One:**
 - **Infection resolving, $\geq$ One:** ⁽⁸⁴⁾
 - Anti-infective therapy completed, arranged, or not indicated
 - Drain removed or drainage manageable
 - *C. difficile* infection and decrease in frequency and severity of diarrhea
 - Infection ruled out
 - **Functional impairment and, One:** ⁽⁴⁾
 - **Discharge to home and, One:** ⁽⁵⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
- **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - **Infection actual or suspected and, Both:**
 - **Finding, $\geq$ One:**
 - Temperature $\geq 100.4^{\circ}F(38.0^{\circ}C)$ ⁽⁸⁵⁾
 - WBC $\geq 12,000/cu.mm(12 \times 10^9/L)$
 - WBC $\leq 4,000/cu.mm(4 \times 10^9/L)$
 - Bands $> 10\%(0.10)$
 - ANC $< 500/cu.mm(500 \times 10^6/L)$ ⁽⁸⁶⁾
 - **Intervention, $\geq$ One:**
 - Anti-infective $\leq 5d$ since initiation ⁽⁸⁷⁾
 - Culture pending $\leq 2d$
 - Drug-induced fever suspected and precipitating drug discontinued $\leq 3d$ ⁽⁸⁸⁾
 - Imaging study $\leq 24h$
 - **Abscess, $\geq$ One:**
 - Intra-abdominal or pelvic abscess and anti-infective $\leq 6d$ since initiation ⁽⁸⁹⁾
 - Abscess drainage requiring drain manipulation $\leq 24h$ ^(90, 91)

- *C. difficile* infection and, **Both**:
 - Worsening frequency and severity of diarrhea
- Intervention, **Both**:
 - Anti-infective (includes PO) ≤ 5d since initiation
 - Continuous IV fluid replacement based on losses ⁽⁹²⁾
- Endocarditis and anti-infective ≤ 6 wks since initiation ⁽⁹³⁾
- Neutropenia and anti-infective and, **Both**: ⁽⁹⁴⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁸⁵⁾
 - ANC < 500/cu.mm(500x10⁶/L) or < 1000/cu.mm(1000x10⁶/L) with documented decline ^(86, 95)
 - Post Critical care ≤ 24h
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
- **Neurology, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **Both**: ⁽²⁾
 - EEG monitoring completed or not indicated
 - Neurological stability ⁽⁹⁶⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Seizure requiring change in medication, dose, or frequency (includes PO) ≤ 2d
 - Continuous EEG ≤ 5d since initiation ⁽⁹⁷⁾
 - Postictal ≤ 24h
 - Post Critical care ≤ 24h
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
- **Obstetrics, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**: ⁽²⁾
 - Endometritis resolving and, ≥ **One**: ⁽⁸⁴⁾
 - Anti-infective regimen established and tolerated
 - Symptoms resolving
 - Preeclampsia with severe features and, **All**: ⁽⁹⁸⁾
 - Blood pressure within acceptable limits ⁽³⁾
 - Laboratory values within acceptable limits ⁽³⁾
 - Neurological stability ⁽⁹⁶⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One**:
 - Postpartum endometritis and anti-infective ⁽⁹⁹⁾
 - Preeclampsia with severe features and, ≥ **One**: ^(98, 100)
 - Antihypertensive (includes PO) ⁽¹⁰¹⁾
 - Magnesium sulfate ⁽¹⁰²⁾
 - Post Critical care ≤ 24h
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
- **Respiratory, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**: ⁽²⁾
 - Clinical stability, ≥ **One**:
 - Oxygenation, ≥ **One**:
 - O₂ sat within acceptable limits ⁽³⁾
 - Oxygen requirement established and, **One**:
 - Home oxygen therapy arranged
 - Post acute facility arranged
 - Chest tube and, **One**: ⁽¹⁰³⁾
 - Heimlich valve
 - Removed and no reaccumulation of air or fluid by chest x-ray (CXR)
 - COPD and, ≥ **One**:
 - O₂ sat within acceptable limits ⁽³⁾

- Arterial $\text{Po}_2 > 60$ mmHg(8.0 kPa) or within acceptable limits ^(3, 104)
- Arterial or venous Pco_2 within acceptable limits ^(3, 104)
- Cystic fibrosis and, **All:**
 - Anti-infective regimen, $\geq$ **One:**
 - Complete
 - Established and tolerated, **One:**
 - Transition to PO anti-infective
 - Vascular access established for IV anti-infective
 - O_2 sat within acceptable limits ⁽³⁾
 - Pulmonary function return to baseline or improving ^(10, 105)
 - Tolerating PO or nutritional route established ^(54, 55)
- **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - O_2 sat 89-91%(0.89-0.91) and $<$ baseline requiring supplemental oxygen and, $\geq$ **One:** ^(10, 106)
 - Chest physiotherapy (CPT) at least 6x/24h, $\leq$ 2d
 - Bronchoscopy performed within last 24h
 - Oxygen therapy adjustments at least 3x/24h and oximetry
 - Unable to tolerate oxygen adjustment $\leq$ 2d
 - Oxygen requirement plateau last 2d and discharge planning $\leq$ 24h ⁽¹⁰⁷⁾
 - Chest tube and, $\geq$ **One:** ⁽¹⁰³⁾
 - Suction, continuous
 - Drainage $\geq$ 250 mL/24h
 - Repositioning within last 24h ⁽¹⁰⁸⁾
 - Water seal within last 24h
 - COPD exacerbation unresolved $\leq$ 3d and, **All:** ^(109, 110)
 - Finding, $\geq$ **One:**
 - O_2 sat $\leq$ 91%(0.91) and $<$ baseline ⁽¹⁰⁾
 - Arterial $\text{Po}_2 \leq 60$ mmHg(8.0 kPa) and $<$ baseline ⁽¹⁰⁴⁾
 - Arterial or venous $\text{Pco}_2 >$ baseline ^(10, 104)
 - Bronchodilator $\geq$ 4x/24h ⁽¹¹¹⁾
 - Corticosteroid (includes PO or inhaled) ⁽¹¹²⁾
 - Oxygenation, **One:** ⁽¹¹³⁾
 - O_2 sat $\geq$ 92%(0.92) or arterial $\text{Po}_2 \geq 60$ mmHg(8.0 kPa) (or baseline) ^(10, 104)
 - O_2 sat $<$ 92%(0.92) or arterial $\text{Po}_2 < 60$ mmHg(8.0 kPa) and $<$ baseline requiring supplemental oxygen ^(10, 104, 106)
 - Oximetry or arterial blood gas ⁽¹¹⁴⁾
 - Cystic fibrosis pulmonary exacerbation unresolved $\leq$ 4d and, **Both:**
 - Airway clearance therapy, **One:** ⁽¹¹⁵⁾
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽¹¹⁶⁾
 - Anti-infective
 - Cystic fibrosis with continued weight loss and initiation or advancement of tube feeding $\leq$ 3d since initiation
 - Post mechanical ventilation or NIPPV $\leq$ 24h and, $\geq$ **One:** ⁽¹¹⁷⁾
 - Oximetry
 - Arterial or venous blood gas ⁽¹⁰⁴⁾
 - Video assisted thoracoscopy performed within last 24h
 - Post Critical care $\leq$ 24h
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
 - **Skin, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁾
 - Skin healing or manageable
 - Skin care regimen established

- Pain controlled or manageable
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Impaired skin integrity ≤ 3d and, ≥ **Two**:
 - Complex wound care ≥ 15 min in duration and performed ≥ 3x/24h ⁽¹¹⁸⁾
 - Parenteral nutrition or tube feeding
 - Analgesic ≥ 3x/24h
 - Anti-infective
 - Dressing change requiring ≥ 2 clinicians
 - Graft integrity at risk or failure
 - Requiring sharp debridement daily
 - Wound on > 2 turning surfaces requiring specialty pressure relieving device
 - Post Critical care ≤ 24h
- **Non-responder**, not clinically stable for discharge and does not meet responder criteria (refer for secondary review)
- **Transplant, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **One**: ⁽²⁾
 - Rejection or graft failure resolving or stabilized
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Rejection or graft failure and immunosuppressant and, ≥ **One**: ^(81, 119)
 - Initiation of new medication ≤ 2d
 - Conversion from IV to PO ≤ 2d ⁽⁸²⁾
 - Medication adjustment (includes PO) ≤ 3d ^(120, 121)
 - Post Critical care ≤ 24h
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria (refer for secondary review)
- **INTERMEDIATE, One**: ⁽¹²²⁾
 - **Responder**, discharge expected today if clinically stable last 12h, **Both**: ⁽²⁾
 - Afebrile
 - Clinical stability, ≥ **One**:
 - Arrhythmia and, ≥ **One**: ^(123, 124)
 - Controlled or resolved ^(123, 124)
 - No syncopal episodes last 12h
 - Oxygenation, **One**:
 - O₂ sat within acceptable limits ⁽³⁾
 - Oxygen requirement established and oxygen therapy arranged
 - Mechanical ventilation or NIPPV and, **All**: ⁽¹²⁵⁾
 - Airway or tracheostomy stable
 - Ventilator management plan established
 - No change in ventilator settings last 2d
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Arrhythmia, ≥ **One**:
 - IV antiarrhythmic and hemodynamically stable
 - Antiarrhythmic conversion from IV to PO, **One**:
 - No proarrhythmic risk ≤ 24h
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d ⁽¹²⁶⁾
 - Hepatic encephalopathy Grade III ≤ 7d since initiation of treatment and, ≥ **One**: ^(57, 62, 127)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽⁶⁴⁾
 - Lactulose (includes PO or PR) ⁽⁶⁵⁾
 - L-ornithine L-aspartate (LOLA) ⁽⁶⁶⁾
 - Neomycin (includes PO) ⁽⁶⁷⁾
 - Rifaximin (includes PO) ⁽⁶⁸⁾
 - Hepatorenal syndrome ≤ 8d since initiation of treatment and, ≥ **One**: ⁽¹²⁸⁾
 - Albumin

- Octreotide and midodrine (includes PO)
- HFNC and respiratory monitoring every 3-4h ^(129, 130)
- Hyperkalemia, **All:** ⁽¹³¹⁾
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, $\geq$ **One:** ⁽¹³¹⁾
 - Peaked T waves
 - PVCs $> 6/\text{min}$
 - Intervention, $\geq$ **One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁴⁵⁾
- Hypokalemia, **All:** ^(47, 48)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs $> 6/\text{min}$
 - Potassium repletion
- Mechanical ventilation and, $\geq$ **One:**
 - Intubated and $\leq 10\text{d}$ since intubation and, $\geq$ **One:**
 - Respiratory interventions every 3-4h ⁽¹³²⁾
 - Weaning $\leq 2\text{d}$ ⁽¹³³⁾
 - Tracheostomy (new) $\leq 7\text{d}$ since placement and discharge planning for post acute care
- NIPPV and, $\geq$ **One:**
 - Respiratory interventions every 3-4h ⁽¹³²⁾
 - Weaning $\leq 2\text{d}$
 - Respiratory plateau $\leq 2\text{d}$ and discharge planning $\leq 24\text{h}$ ⁽¹⁰⁷⁾
- Oxygen $\geq 40\%(0.40)$ and, $\geq$ **One:** ⁽¹⁰⁶⁾
 - Chest physiotherapy (CPT) at least 6x/24h, $\leq 2\text{d}$
 - COPD $\leq 2\text{d}$ and, **Both:**
 - Bronchodilator $\geq 4\text{x}/24\text{h}$ ⁽¹¹¹⁾
 - Corticosteroid (includes PO or inhaled)
 - Bronchoscopy performed within last 24h
 - Post transplant, lung or heart-lung $\leq 4\text{d}$
 - Oxygen therapy adjustment at least 3x/24h and oximetry
- Pericardial effusion and high risk for cardiac tamponade $\leq 24\text{h}$ and, **Both:** ^(134, 135, 136)
 - Asymptomatic
 - Effusion size $\geq 1\text{ cm}$ ⁽¹³⁷⁾
- Seizure and, **Both:**
 - Activity resolved
 - Anticonvulsant, continuous
- VAD and, $\geq$ **One:** ^(138, 139)
 - Anticoagulation, $\geq$ **One:** ^(140, 141, 142)
 - Direct thrombin inhibitor
 - GP IIb or IIIa inhibitor
 - Heparin
 - Thrombolytic
 - Bleeding actual or suspected and, $\geq$ **One:**
 - Hct or Hb monitoring at least 3x/24h, $\leq 3\text{d}$
 - Volume expander or blood product ⁽⁷⁶⁾
 - Inadequate nutrition and, $\geq$ **One:**
 - Calorie count $\leq 3\text{d}$
 - Requiring daily tube feeding adjustment
 - TPN initiation $\leq 24\text{h}$
 - Infection actual or suspected and, **Both:**
 - Finding, $\geq$ **One:**
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽⁸⁵⁾

- WBC $\geq 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$
- WBC $\leq 4,000/\text{cu.mm}(4 \times 10^9/\text{L})$
- Bands $> 10\%(0.10)$
- ANC $< 500/\text{cu.mm}(500 \times 10^6/\text{L})$ ⁽⁸⁶⁾
- Anti-infective
- Post acute facility unavailable and, $\geq$ **One**:
 - Functional impairment requiring therapy ⁽⁴⁾
 - Inadequate social or family support
 - Patient or caregiver unable to demonstrate functional ability to maintain and operate VAD
- Post Critical care $\leq 3\text{d}$
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria for Intermediate or Acute (refer for secondary review)
- **CRITICAL, One:** ⁽¹⁴³⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One**:
 - Acute humoral rejection and apheresis ^(144, 145)
 - AKI and, **Both:** ⁽⁷²⁾
 - Finding, $\geq$ **One**:
 - Potassium $> 6.0 \text{ mEq/L}(6.0 \text{ mmol/L})$ and, $\geq$ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹⁴⁶⁾
 - Widening QRS
 - Sodium, **One**:
 - $< 120 \text{ mEq/L}(120 \text{ mmol/L})$ and volume overload
 - $> 160 \text{ mEq/L}(160 \text{ mmol/L})$
 - Intervention, $\geq$ **One**:
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁴⁵⁾
 - Diuretic, continuous
 - Dialysis, **One**:
 - Hemodialysis $\leq 7\text{d}$ since initiation
 - Peritoneal dialysis $\leq 7\text{d}$ since initiation
 - IV fluid resuscitation $\leq 2\text{d}$ ⁽¹⁴⁷⁾
 - Anticonvulsant and, $\geq$ **One**:
 - Recurrent seizure ⁽¹⁴⁸⁾
 - Refractory seizure ⁽¹⁴⁸⁾
- Arrhythmia, $\geq$ **One**:
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
- Symptomatic recurrent or refractory arrhythmia requiring antiarrhythmic and, $\geq$ **One**: ⁽¹⁴⁹⁾
 - Chest pain
- Heart failure and, $\geq$ **One**:
 - Arterial $\text{Po}_2 < 56 \text{ mmHg}(7.4 \text{ kPa})$
 - $\text{O}_2 \text{ sat} < 89\%(0.89)$ and $<$ baseline
- Hemodynamic instability, $\geq$ **One**:
 - Heart rate $> 120/\text{min}$, sustained ⁽¹⁵⁰⁾
- Systolic blood pressure, $\geq$ **One**:
 - $< 90 \text{ mmHg}$
 - $< 110 \text{ mmHg}$ with chronic hypertension
 - $> 30 \text{ mmHg}$ decrease from baseline ⁽¹⁰⁾
 - Labile ⁽¹⁵¹⁾

- MAP < 65 mmHg
- CRRT ⁽¹⁵²⁾
- Cerebral edema and, ≥ **One**:
 - ICP monitoring (includes OR placement)
 - Corticosteroid
 - Hyperosmolar therapy, ≥ **One**: ⁽¹⁵³⁾
 - Mannitol
 - Hypertonic saline ≥ 3%(0.03) solution
- DIC and blood product transfusion ⁽¹⁵⁴⁾
- ECMO or ECLS ⁽¹⁵⁵⁾
- Fulminant *C. difficile* infection and, **Both**: ⁽¹⁵⁶⁾
 - Ileus ⁽¹⁵⁷⁾
 - IV fluid resuscitation ≤ 2d ^(147, 158)
- Heart failure and, **Both**:
 - Finding, ≥ **One**:
 - Arterial Po₂ < 56 mmHg(7.4 kPa)
 - O₂ sat < 89%(0.89) and < baseline ⁽¹⁰⁾
 - Diuretic every 1-2h or continuous
- Hemodynamic monitoring, invasive ≤ 24h ⁽¹⁵⁹⁾
- Hepatic encephalopathy Grade IV ≤ 7d since initiation of treatment and, ≥ **One**: ^(57, 62, 127)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽⁶⁴⁾
 - Lactulose (includes PO or PR) ⁽⁶⁵⁾
 - L-ornithine L-aspartate (LOLA) ⁽⁶⁶⁾
 - Neomycin (includes PO) ⁽⁶⁷⁾
 - Rifaximin (includes PO) ⁽⁶⁸⁾
- Hepatorenal syndrome ≤ 8d since initiation of treatment and, ≥ **One**: ⁽¹²⁸⁾
 - CRRT ⁽¹⁵²⁾
 - Hemodialysis or peritoneal dialysis 2d since initiation
 - Norepinephrine
- HFNC and respiratory monitoring every 1-2h ^(129, 130)
- Hypercalcemia, **Both**: ^(37, 38)
 - Calcium ≥ 14.0 mg/dL(3.5 mmol/L)
 - Intervention, ≥ **One**:
 - CRRT ⁽¹⁵²⁾
 - Dialysis, hemodialysis or peritoneal
 - Treatment, **Both**: ⁽⁴⁰⁾
 - Calcitonin
 - Saline hydration ^(40, 41)
- Hypocalcemia, **Both**: ^(38, 42)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)
 - Calcium repletion
- Hyperkalemia, **Both**: ⁽¹³¹⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹⁴⁶⁾
 - Widening QRS
 - Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - Dialysis, **One**:
 - CRRT ⁽¹⁵²⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽⁴⁵⁾

- Hypokalemia, **All:** ^(47, 48, 160)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
- Electrocardiogram (ECG) findings, ≥ **One:** ⁽¹⁶⁰⁾
 - Multifocal PVCs
 - R on T phenomenon ⁽¹⁶¹⁾
 - T and U wave fusion ⁽¹⁶²⁾
 - U wave ⁽¹⁶³⁾
- Potassium repletion ⁽⁴⁸⁾
- Hypomagnesemia, **Both:**
 - Magnesium < 1.0 mg/dL(0.41 mmol/L)
 - Magnesium repletion
- Hypernatremia, **Both:** ⁽⁵⁰⁾
 - Sodium > 160 mEq/L(160 mmol/L)
- Intervention, ≥ **One:**
 - CRRT ⁽¹⁵²⁾
 - Dialysis, hemodialysis or peritoneal
 - IV fluid ⁽⁴¹⁾
- Hyponatremia, **Both:**
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(164, 165)
- Intervention, ≥ **One:**
 - Conivaptan or tolvaptan (includes PO) ⁽¹⁶⁶⁾
 - Diuretic ≥ 2x/24h
 - Fluid restriction
 - Sodium chloride 3%(0.03) ⁽⁵²⁾
- Hypophosphatemia, **Both:** ⁽⁵³⁾
 - Phosphate < 1.0 mg/dL(0.32 mmol/L)
 - Phosphate repletion
- IABP ^(167, 168)
- Mechanical ventilation, discharge planning and, ≥ **One:** ⁽¹⁶⁹⁾
 - FiO₂ > 50%(0.50) and > baseline ^(10, 106)
 - PEEP > 10 cm H₂O and > baseline ⁽¹⁰⁾
 - Respiratory interventions every 1-2h ⁽¹³²⁾
 - Active weaning < 3 attempts ^(169, 170)
 - Failed weaning ≥ 3 attempts and tracheostomy placed ≤ 24h ⁽¹⁷¹⁾
- NIPPV and, ≥ **One:** ⁽¹²⁵⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽¹³²⁾
- Pericardial effusion and, ≥ **One:** ^(172, 173)
 - Symptomatic ≤ 24h, ≥ **One:** ⁽¹⁷⁴⁾
 - Dyspnea
 - Jugular venous distention
 - Orthopnea
 - Insertion of pericardial drain performed ≤ 24h ⁽¹⁷⁵⁾
 - Pericardiocentesis performed ≤ 24h ⁽¹⁷⁵⁾
- Post embolization ≤ 24h
- Thrombolytic ⁽¹⁷⁶⁾
- Toxic megacolon and, ≥ **One:** ⁽¹⁷⁷⁾
 - Blood product transfusion
 - Colonic decompression
 - IV fluid resuscitation ≤ 2d ⁽¹⁴⁷⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- VAD, percutaneous ^(139, 178)
- VAD and, ≥ **One:** ^(138, 179)

- Blood pressure, **≥ One:**
 - < baseline and inadequate perfusion ⁽¹⁸⁰⁾
 - MAP < 70 mmHg and continuous flow device ⁽¹⁸¹⁾
 - Systolic < 90 mmHg and pulsatile flow device
- Decreased level of consciousness
- Decreased VAD flows and inadequate perfusion ⁽¹⁸⁰⁾
- O₂ sat < 89%(0.89)
- **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria for Critical, Intermediate, or Acute (refer for secondary review)

Discharge, Level of Care, One: ⁽¹⁸²⁾● **HOME, Both:**● Level of care appropriateness, **Both:**

- Home environment safe and accessible ⁽¹⁸³⁾
- Patient or caregiver demonstrates ability to manage care

● Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽¹⁸⁴⁾
- Medication reconciliation ⁽¹⁸⁵⁾
- Follow-up care planned ⁽¹⁸⁶⁾
- Identify and address transportation needs

● **HOME CARE, Both:**● Level of care appropriateness, **All:**

- Home environment safe and accessible ⁽¹⁸³⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(187, 188)

● Skilled services, **≥ One:** ⁽¹⁸⁹⁾

- Skilled nursing ⁽¹⁹⁰⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽¹⁹¹⁾

● Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽¹⁸⁴⁾
- Medication reconciliation ⁽¹⁸⁵⁾
- Follow-up care planned
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**● Level of care appropriateness, **One:**

- Support systems available ⁽¹⁹²⁾

● Skilled treatment, **≥ One:**

- Clinical assessment ⁽¹⁹³⁾
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation ⁽¹⁹⁴⁾

● **SKILLED MEDICAL OR THERAPY, Both:**● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight **≥ 1x/wk**
- Treatment precluded in a lower level

● Services, **≥ One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽¹⁹⁰⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(195, 196)
- Functional impairments requiring at least supervision ⁽¹⁹⁷⁾
- Able to tolerate 1h/d of skilled therapy **≥ 5d/wk**

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽¹⁸⁵⁾

- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 2x/wk$
- Treatment precluded in a lower level

● Services, $\geq$ **One:**

- Medical and skilled nursing interventions or assessment 4h/24h ⁽¹⁹⁰⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(195, 196)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁹⁷⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 1 therapy discipline

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽¹⁸⁵⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE REHAB, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- Treatment precluded in a lower level

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(195, 196)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁹⁷⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽¹⁸⁵⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(195, 196)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁹⁷⁾
- Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽¹⁹⁸⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽¹⁸⁵⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

● Level of care appropriateness, **Both:**

- Treatment precluded in a lower level

● In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**

- Primary condition or illness and treatment of active comorbid condition(s) ⁽¹⁹⁹⁾
- Ventilator dependent $\geq 6h/d$ and weaning planned ^(200, 201)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽²⁰²⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽¹⁸⁵⁾
- Obtain and complete forms for facility

- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Notes:**1:**

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

2:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

3:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

4:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

5:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

6:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs

- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

7:

An outpatient anticoagulation regimen may include low molecular weight heparin (LMWH), fondaparinux, warfarin, or a direct oral anticoagulant (DOAC). For patients who will be treated with warfarin, initiation of treatment must overlap with parenteral anticoagulation until a therapeutic international normalized ratio (INR) is achieved; therefore, some patients could be discharged on parenteral anticoagulant along with warfarin if discharged prior to completion of overlap therapy. Dabigatran and edoxaban, both DOACs, require 5 to 10 days of pretreatment with parenteral anticoagulation; therefore, these patients may be discharged on a parenteral anticoagulation with the plan to switch to dabigatran or edoxaban after pretreatment is complete (Ortel et al., Blood Adv 2020, 4: 4693-738). This criterion also includes patients with confirmed or a history of heparin-induced thrombocytopenia (HIT) who have achieved therapeutic anticoagulation and whose outpatient regimen has been established.

8:

Blood transfusion practice guidelines are intended to improve clinical outcomes while avoiding unnecessary transfusions that may not be clinically indicated. Recent high-quality evidence has resulted in guideline updates recommending lower hemoglobin transfusion thresholds for most hospitalized adult and pediatric patients. While individual institutional practice patterns may vary, the American Association of Blood Banks (AABB) recommendation for hemodynamically stable patients, including critically ill patients, is a restrictive transfusion threshold for a hemoglobin of less than 7 g/dL for most patients. A hemoglobin threshold of 7.5 g/dL is recommended for transfusion if patients are undergoing cardiac surgery and a threshold of 8 g/dL if patients are undergoing orthopedic surgery or have pre-existing cardiac disease. There are exceptions to these parameters including patients with acute coronary syndrome, severe thrombocytopenia, and chronic transfusion-dependent anemia (Carson et al., Jama 2023: E1-E11; Chegondi et al., Transfus Med Hemother 2016, 43: 297-301).

9:

Patients with severe anemia may exhibit exertional dyspnea (e.g., shortness of breath while walking up a flight of stairs) due to lack of circulating oxygen-carrying red blood cells to muscles. This may require hospitalization for blood transfusion depending on thresholds and possible need for supplemental oxygen (Yui and Brodsky, Hematol Oncol Clin North Am 2022, 36: 325-39; Long and Koyfman, Emerg Med Clin North Am 2018, 36: 609-30; DeVos and Jacobson, Emerg Med Clin North Am 2016, 34: 129-49).

10:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

11:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

12:

Orthostatic hypotension (Juraschek et al., JAMA Intern Med 2017, 177: 1316-23).

13:

Syncope is the transient loss of consciousness caused by diminished cerebral blood flow, identified as brief, with spontaneous onset and recovery (Brignole et al., Eur Heart J 2018, 39: 1883-948; Shen et al., Journal of the American College of Cardiology 2017, 70: e39-e110).

14:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

15:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

16:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

17:

Hemolytic anemia is caused by various intrinsic and extrinsic factors leading to abnormal, premature destruction of red blood cells. Hemolysis may lead to emergent complications, such as acute thrombophilia, heart failure, and/or chronic kidney disease, which may require hospitalization (Verbillion and Dupre, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:1462-79; Robertson et al., J Emerg Med 2017, 53: 202-11).

18:

In the setting of warm antibody autoimmune hemolytic anemia, immediate oral or intravenous administration of corticosteroids is indicated. Administration is continued until a hematocrit of greater than 30%(0.30) or a hemoglobin level of greater than 10.0 g/dL(100 g/L) is achieved. Other types of hemolytic anemia are refractory to corticosteroid treatment and may require additional interventions (Jäger et al., Blood Rev 2020, 41: 100648; Brodsky, N Engl J Med 2019, 381: 647-54).

19:

Intravenous immunoglobulin (IVIG) is a blood product which contains immunoglobulins extracted, then pooled, from the plasma of thousands of blood donors. Its administration carries the same risks as those of any blood product transfusion, including headache, backache, chills, nausea, muscle pain, and, although rare, anaphylaxis. IVIG is used for a wide range of disorders, and its therapeutic mechanism varies based on the dose given and the patient's disease. IVIG is used for the treatment of diseases in neurology, hematology, immunology, dermatology, and nephrology.

20:

Heparin-induced thrombocytopenia (HIT) results from sensitization to the heparin molecule. Usually, it occurs between 5 to 10 days after the current heparin exposure but may occur as early as minutes to hours after exposure. HIT is a greater than 50%(0.50) serial reduction in a patient's baseline platelet count. A direct thrombin inhibitor (DTI) (e.g., argatroban, bivalirudin) is the initial treatment for patients with active HIT, a history of HIT, or those with a strong clinical suspicion of HIT who require anticoagulation. Although the American Society of Hematology (ASH) guidelines suggest the use of fondaparinux or direct oral anticoagulants (DOACs), the recommendation is conditional and based on very low certainty of evidence. In addition, these agents are considered off-label use in the treatment of HIT since they have yet to be approved by the U.S. Food and Drug Administration (FDA) for this condition. For patients who require long-term anticoagulation, warfarin should not be initiated until platelet count recovers (e.g., platelet count greater than $150 \times 10^9/L$). Additionally, a 5-day overlap with warfarin and DTI therapy is recommended (Cuker et al., Blood Adv 2018, 2: 3360-92; Guyatt et al., Chest 2012, 141: 48S-52S).

21:

Instruction: This criterion applies to patients with actual, suspected, or a history of heparin-induced thrombocytopenia (HIT).

22:

In a large cohort study of over 4,000 patients with acute pancreatitis pain, about 80%(0.80) received opioids during the initial stages of hospitalization. Increased opioid use was found to be associated with a longer length of hospitalization, with about 10%(0.10) of patients continuing to use opioids after discharge. This continued use of opioids may reflect in chronic pancreatitis management (Wu et al., JAMA Netw Open 2019, 2: e191827).

23:

Pain has been reported in up to 71% of hospitalized patients. Despite risks of opioid use, more than half of hospitalized patients received a dose of opioid during admission. Opioid use in hospitalized patients has been associated with adverse outcomes, such as nausea, constipation, urinary retention, falls, delirium, sedation, physical dependence, addiction, respiratory depression, and death. Patients most at risk for opioid related adverse events include age over 65 years, renal insufficiency, hepatic insufficiency, chronic respiratory failure, and receipt of other central nervous system depressant medications. Guidelines that determine safe opioid practice for patients hospitalized in the acute, general medical setting do not exist. In response, the Society of Hospital Medicine developed a consensus statement regarding safe opioid prescribing practices for hospitalized patients outside of surgical, oncological, and intensive care settings. These include (Herzig et al., J Hosp Med 2018, 13: 263-71; Herzig et al., J Hosp Med 2018, 13: 256-62):

- Trial of non-opioid therapy prior to use of opioids, including non-pharmacologic treatments
- Non-opioid analgesics (acetaminophen or non-steroidal anti-inflammatory medications) have been shown to be as effective as opioids with less side effects
- Restrict opioid use to severe pain, pain that has not responded to non-opioid therapy, or patients in whom non-opioid therapy is contraindicated
- Use short-acting opioids and/or avoid use of long-acting/extended-release opioids
- Limit treatment to 14 days (including period of hospitalization for discharge planning)

24:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

25:

The American Society of Hematology (ASH) guidelines for management of venous thromboembolism (VTE) recommend offering home treatment over hospital treatment for patients with pulmonary embolism (PE) with low risk for complications and uncomplicated deep vein thrombosis (DVT). However, this recommendation does not apply to individuals who have inadequate home support, a lack of access to medication, or a history of poor compliance (Ortel et al., Blood Adv 2020, 4: 4693-738). Patients with these risk factors may require inpatient admission while receiving parenteral anticoagulation (e.g., fondaparinux, low molecular weight heparin, unfractionated heparin) during warfarin bridging therapy or pre-treatment therapy for dabigatran or edoxaban.

26:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting
- Skilled nursing services unavailable (e.g., Home Care)

27:

Instruction: Patients who have not achieved a therapeutic INR on 2 consecutive days and continue on LMWH or

fondaparinux will meet this criterion.

According to the American Society of Hematology (ASH), when warfarin is being initiated, it should be overlapped with parenteral anticoagulation (e.g., fondaparinux, low molecular weight heparin (LMWH)) until a therapeutic international normalized ratio (INR) of 2.0-3.0 is achieved (Ortel et al., Blood Adv 2020, 4: 4693-738).

28:

Anticoagulation may include the following:

- Low molecular weight heparin (LMWH)
- Unfractionated heparin (UFH), continuous
- Fondaparinux

29:

Patients treated with dabigatran or edoxaban, both direct oral anticoagulants (DOACs), require at least 5 days of parenteral anticoagulation prior to initiation (Ortel et al., Blood Adv 2020, 4: 4693-738).

30:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

31:

Patients with heart failure exacerbation who are slow to respond to diuretic therapy may benefit from evaluation by cardiologist or heart failure specialist and may require further assessment or additional treatments, such as (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011):

- Higher dose of diuretics or addition of a second diuretic
- Invasive hemodynamic monitoring through right heart catheterization or pulmonary artery catheter
- Re-evaluation of comorbidities and/or alternate diagnosis such as ischemic heart disease, lung disease, or thyroid disease

32:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation 2015).

33:

Inotropes (e.g., dopamine, dobutamine, and milrinone) can be used in patients with severe systolic dysfunction resulting in hypotension and low cardiac output. These medications can be effective in treating patients who are unable to tolerate or do not respond to vasodilators and diuretics. The goal of treatment is to maintain systemic perfusion and improve end-organ hypoperfusion. Temporary use of inotropes is recommended for hospitalized patients in cardiogenic shock, and more prolonged use as a bridge therapy may be indicated in advanced heart failure (HF) patients (stage D) who are awaiting mechanical circulatory support (MCS) or heart transplant. They may also be indicated as a palliative therapy for those unable to undergo MCS or transplantation. Patients on inotropes require close monitoring as they are either unstable or have advanced disease, and inotropes themselves

carry risks such as arrhythmias (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

34:

Instruction: Infrequent titration of low dose inotropes can be provided at the Acute level of care. A less intensive level of care may be appropriate for patients requiring low dose inotropes on a long-term basis.

35:

Initiation refers to the first time an intervention is utilized or administered during this hospitalization.

36:

Instruction: PO Vasodilators can be given at the Acute level of care if the patient is hemodynamically stable.

37:

Hypercalcemia (Mousseaux et al., Ann Intensive Care 2019, 9: 133; Turner, Clin Med (Lond) 2017, 17: 270-3; Maier and Levine, J Intensive Care Med 2015, 30: 235-52).

38:

Calcium levels within criteria represent total serum calcium.

39:

Bisphosphonates and corticosteroids are more commonly used in the treatment of hypercalcemia related to malignancy.

40:

In hypercalcemia, saline diuresis is administered to maintain an output of 200 mL per hour.

41:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

42:

Abnormalities of calcium can have significant impact on morbidity and mortality due to its crucial role in neuromuscular and cardiac function. Patients who present with symptoms usually have an abrupt decline in ionized calcium.

Common symptoms of hypocalcemia include:

- Perioral numbness
- Carpopedal spasm of the hands or feet
- Flaccid paralysis
- Muscle weakness
- Paresthesia
- Tetany

Symptomatic hypocalcemia requires treatment with intravenous calcium (Turner et al., Endocr Connect 2019, 8: X1).

43:

In patients with no underlying heart disease, occasional, nonconsecutive premature ventricular contractions (PVCs) less than 6/min are common and not considered a concerning finding. Three or more consecutive PVCs at a rate of 100 bpm or more that terminate spontaneously within 30 seconds are considered nonsustained ventricular tachycardia (Marine, Card Electrophysiol Clin 2016, 8: 525-43; Buxton et al., Circulation 2006, 114: 2534-70).

44:

Following an initial 12-lead electrocardiogram (ECG) demonstrating no abnormal findings in the setting of hyperkalemia or hypokalemia, continued monitoring may be accomplished by telemetry (Bridgeman et al., Nephrol Dial Transplant 2019, 34: iii45-iii50).

45:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

46:

Examples of potassium binding agents include but are not limited to:

- Sodium polystyrene sulfonate (Kayexalate® or Kionex®)
- Sodium zirconium cyclosilicate (Lokelma®)
- Patiomer (Veltassa®)

47:

Hypokalemia rarely occurs in isolation and is usually secondary to a disease process or overuse of diuretics. Criteria should be reviewed for the underlying cause of the hypokalemia (e.g., diabetic ketoacidosis, gastroenteritis) to ensure the patient is managed appropriately.

48:

When a patient with hypokalemia is not responding to potassium repletion, the magnesium level should be checked since hypomagnesemia could be contributing to the refractory hypokalemia.

49:

Hypermagnesemia (Dalia et al., J Cardiothorac Vasc Anesth 2019, 33: 1722-30)

50:

Hypernatremia (Muhsin and Mount, Best Pract Res Clin Endocrinol Metab 2016, 30: 189-203)

51:

Hyponatremia (Seay et al., Am J Kidney Dis 2020, 75: 272-86)

52:

Hypovolemic hyponatremia may be treated with sodium chloride 0.9% (0.009). For severe hyponatremia, treatment with an infusion of sodium chloride 3% (0.03) may be required. Rapid correction of serum sodium can cause spastic quadriparesis or quadriplegia, pseudobulbar palsy, dysarthria, mutism, pupillary or oculomotor abnormalities, stupor, coma, ataxia, dystonia, osmotic demyelinating syndrome, or even death (Weismann et al., Endocr Connect 2016:).

53:

Hypophosphatemia (Megapanou et al., Drug Saf 2020, 43: 197-210)

54:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

55:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

56:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

57:

Hepatic encephalopathy (HE) is a complication of acute or chronic liver disease that is primarily characterized by changes in personality, consciousness, cognition, and motor function. West Haven criteria for grading HE includes (Montrief et al., Am J Emerg Med 2019, 37: 329-37; Weissenborn, Drugs 2019, 79: 5-9; Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47):

- Grade I - Lack of awareness, anxiety, short attention span, euphoria, inability to add/subtract
- Grade II - Disorientation for time, inappropriate behavior, lethargy/apathy, obvious personality change, dyspraxia, or asterixis
- Grade III - Confusion, bizarre behavior, gross disorientation, responsive to verbal stimuli, somnolence to semi-stupor
- Grade IV - Coma

58:

Patients who may require continued stay with ascites include those who have increasing or unchanged abdominal girth or weight, as well as patients who are improved but not clinically stable for discharge because they have had a subtherapeutic response to treatment.

59:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

60:

The following list of generic medications may be prescribed for nausea and/or vomiting: ondansetron, promethazine, metoclopramide, prochlorperazine, meclizine, lorazepam, hydroxyzine, scopolamine, dimenhydrinate or diphenhydramine, haloperidol, trimethobenzamide, chlorpromazine, droperidol, perphenazine, pyridoxine, or doxylamine/pyridoxine.

61:

Serotonin 5-HT₃ receptor antagonists are typically given to help prevent nausea and vomiting. Examples of these drugs include: ondansetron, dolasetron, granisetron, and palonosetron.

62:

While an elevated ammonia level is the classic laboratory finding used to diagnose hepatic encephalopathy, serial ammonia measurements are inferior to clinical assessment in gauging improvement or deterioration in a patient receiving treatment (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47).

63:

Instruction: Patients with end-stage liver disease are generally on chronic outpatient lactulose, rifaximin, or

neomycin at home to control symptoms of hepatic encephalopathy (HE). Admission to a skilled nursing facility (SNF) may be appropriate to initiate or adjust the dosage of these medications due to worsening hepatic encephalopathy.

64:

Oral branched-chain amino acid (BCAA) enriched formulations may improve the manifestations of occult or minimal episodic hepatic encephalopathy (HE). Intravenous administration is ineffective (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47; J Hepatol 2014, 61: 642-59). Although BCAA has a beneficial effect on HE, there is no evidence regarding an effect on mortality, quality of life, or nutritional parameters (Gluud et al., Cochrane Database Syst Rev 2017, 5: CD001939).

65:

Lactulose is recommended by the American Association for the Study of Liver Diseases/European Association for the Study of the Liver guidelines as the first-line treatment for episodes of hepatic encephalopathy. It is a nonabsorbable substance used to treat an elevation in blood ammonia (NH_3) and reduces plasma ammonia concentration by converting soluble ammonia into insoluble ammonia in the colon. The laxative effect of lactulose results in the expulsion of trapped ammonium in the form of diarrhea. Lactulose can be administered orally, via a nasogastric tube, or as a retention enema (Swaminathan et al., Hepat Med 2018, 10: 1-11; J Hepatol 2014, 61: 642-59).

66:

Evidence has shown that L-ornithine L-aspartate (LOLA) is effective in lowering ammonia and improving mental status in patients with cirrhosis and hepatic encephalopathy (HE), with the oral form of LOLA being especially beneficial (Butterworth et al., J Clin Exp Hepatol 2018, 8: 301-13). In a 2017 randomized, controlled trial, patients with persistent HE showed improvement from intravenous LOLA administration, whereas oral supplementation was determined to be ineffective (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47). A recent systematic review reassessed both oral and intravenous (IV) LOLA for the treatment of low-grade or minimal hepatic encephalopathy (MHE) and high grade or overt hepatic encephalopathy (OHE). The relative efficacy of LOLA depended on the type of HE as well as the route of administration. Patients with MHE, defined as patients with cirrhosis and alterations in mental status diagnosed by psychometric testing, appear to benefit more from the oral administration of LOLA than patients with high grade overt HE (Weissenborn, Drugs 2019, 79: 5-9; Butterworth et al., J Clin Exp Hepatol 2018, 8: 301-13).

67:

The renal and ototoxicity of neomycin have rendered this medication nearly obsolete in the treatment of hepatic encephalopathy (Tabbers et al., J Pediatr Gastroenterol Nutr 2014, 58: 258-74).

68:

Rifaximin is an anti-infective recommended by evidence-based clinical guidelines as an effective add-on therapy to lactulose for the treatment of overt hepatic encephalopathy (OHE). It has been shown to reduce healthcare resource utilization over the long term by lessening OHE recurrences and rehospitalization (Hudson and Schuchmann, European journal of gastroenterology & hepatology 2019, 31: 434). In a systematic review comparing the effectiveness and safety of interventions in patients with OHE, Rifaximin showed the greatest reduction in blood ammonia concentration (Zhu et al., Aliment Pharmacol Ther 2015, 41: 624-35).

69:

An ileus is a blockage, partial or complete, or paralysis of the bowel that prevents the passage of intestinal contents from moving forward. It is more common in the small bowel, as opposed to large bowel. It is often the result of surgical adhesions or hernia. Small bowel ileus symptoms typically present as acute and severe, with complaints of nausea and vomiting, cramps, bloating, and the retention of stool and flatus. It can often be treated medically. Large bowel ileus usually has a milder presentation and often is due to cancer. It may also be the result of adhesions or stenoses from recurrent diverticulitis or, less commonly, is associated with a sigmoid volvulus or hernia (Vilz et al., Dtsch Arztebl Int 2017, 114: 508-18).

70:

Maintenance fluid rates can be determined using the 4-2-1 calculation formula, which is based on ideal body weight

in kilograms. To convert pounds to kilograms:

- weight (lbs.) divided by 2.2 = weight (kg)

To calculate the intravenous fluid rate:

- For the first 10 kg at 4 mL/kg/h = 40 mL/h
- For the second 10 kg at 2 mL/kg/h = 20 mL/h
- For each kg over 20 kg = 1 mL/h up to 125 mL/h

Instruction: Maintenance fluid rates are the minimum amount the patient needs to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss.

71:

Proton pump inhibitors (PPI) reduce rates of rebleeding and mortality in patients with a bleeding ulcer. Current guidelines recommend continuous or intermittent high-dose PPI, defined as at least 80 mg per day for 3 days, following endoscopic hemostasis in patients with a bleeding ulcer (Laine et al., Am J Gastroenterol 2021, 116: 899-917).

72:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.

AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.

AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

73:

In this instance, oliguria is not the result of an outlet obstruction.

74:

Urine output less than 0.5 mL/kg/h may be considered oliguria.

75:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

76:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.

Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

77:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in

volume-responsive patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Pistolesi et al., J Nephrol 2018, 31: 797-812; Joannidis et al., Intensive Care Med 2017, 43: 730-49; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

78:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

79:

Dialysis includes hemodialysis or peritoneal dialysis. Both modalities are equally effective and should be provided based on patient needs and device availability (Liu et al., Cochrane Database Syst Rev 2017, 12: Cd011457).

80:

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

81:

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

82:

Instruction: If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

83:

Prior to the initiation of immunosuppressants for systemic diseases, which adversely impacts kidney function, a renal biopsy may be necessary.

84:

Resolution of infection is indicated by a decrease in temperature, a normalizing of white blood cells and other lab values, and improving imaging studies. If there is a wound present, signs of infection should be resolving (e.g., decrease in wound drainage, less pain at site).

85:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

86:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than 500/cu. mm($500 \times 10^6/L$) are severely neutropenic and have a very limited immunologic response to infection.

To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) x WBC.

Examples:

ANC calculation using a percentage of total WBC count:

WBC count: 1,500/cu.mm($1.5 \times 10^9/L$)

Segs: 20%(0.20) of the WBCs

Bands: 2%(0.02) of the WBCs

$(20\% + 2\%(0.20 + 0.02)) \times 1,500/\text{cu.mm}(1.5 \times 10^9/L)$

$= 330/\text{cu.mm}(330 \times 10^6/L)$

Normal range: 1,500-8,000/cu.mm($1.5-8.0 \times 10^9/L$)

An ANC of less than 500/cu.mm($500 \times 10^6/L$) is considered severely neutropenic.

ANC Calculation using the above numbers as whole numbers:

WBC count: 1.5

Segs: 0.2

Bands: 0.02

$(0.2 + 0.02) \times 1.5 = 0.33$

Normal range: 1.5 to 8.0

An ANC of less than 0.5 is considered severely neutropenic.

87:

In most patients hospitalized with infection, clinical improvement would be expected to occur within 2 days. For patients who are deteriorating or not responding to treatment, it may be necessary to broaden anti-infective coverage while awaiting the results of additional diagnostic tests. In addition, consultation by an infectious disease specialist and further diagnostic tests may be valuable to evaluate reasons for the lack of improvement.

88:

A drug-induced fever is a non-infectious febrile response following the administration of a drug that resolves when the drug is discontinued. The patient is often well-appearing and demonstrates resolution of the underlying disease process, despite the presence of a fever. A drug-induced fever should be suspected when the cause of the fever remains unknown, despite a full diagnostic work-up, or when the patient is receiving a drug frequently associated with fever. The most common agents triggering fever include anticonvulsants, anti-infectives, cardiac medications, allopurinol, and heparin.

The following drugs are also capable of inducing hyperthermia (Walter and Carraretto, Journal of the Intensive Care Society 2015):

- Serotonin-noradrenalin reuptake inhibitors
- Selective serotonin reuptake inhibitors
- Monoamine oxidase inhibitors
- Tricyclic antidepressants
- Bupropion
- Opioids
- Central nervous system stimulants
- Anticholinergics
- Antipsychotics
- Antispasmodics
- Sympathomimetic agents

Treatment involves discontinuing medications that are unnecessary, recently initiated, or highly associated with inducing fever. Resolution of the fever should occur within 48 to 72 hours of discontinuing the precipitating agent (Patel and Gallagher, Pharmacotherapy 2010, 30: 57-69; Eyer and Zilker, Crit Care 2007, 11: 236).

89:

Patients with an intra-abdominal or a pelvic abscess typically present with a sudden onset of abdominal pain along with gastrointestinal symptoms (e.g., nausea, vomiting, diarrhea). A fever may or may not be present. Computed tomography (CT) is the preferred diagnostic imaging technique due to high accuracy. The size and location of the abscess will determine whether treatment with intravenous anti-infectives alone is sufficient (e.g., smaller fluid collections) and/or when drainage may be appropriate. If symptoms worsen or do not improve with antibiotics alone, drainage may subsequently be needed (DeCesare et al., Am Surg 2021, 87: 120-4; Thompson and Hrabe, Gastroenterol Clin North Am 2021, 50: 475-88; Workowski et al., MMWR Recomm Rep 2021, 70: 1-187).

90:

Occasionally, the abscess may evolve and require additional drain placement or require drain repositioning or upsizing (Thompson and Hrabec, Gastroenterol Clin North Am 2021, 50: 475-88).

91:

Drain manipulation includes:

- Placing an additional drain
- Repositioning existing drain
- Placing a larger drain

92:

Fluid is replaced based on measured and insensible losses. Examples of measurable fluid loss include:

- Urine
- Nasogastric
- Wound
- Ostomy drainage
- Diarrhea
- Vomiting
- Fluid sequestration

Examples of insensible loss include:

- Perspiration
- Breathing.

The intent of the criteria is to address patients who have excessive fluid loss due to diarrhea and/or inability to take adequate oral fluids to replace fluid loss. If the losses are due to diarrhea, there is no expectation that the order for intravenous fluids will specify measurement of the diarrhea, replacement on a cubic centimeter (cc) for cc basis, or any titration of the fluid rate based on the number of cc's lost. Continuous intravenous fluid should be ordered in a clinically significant amount that indicates a need for replacement along with documentation of excessive diarrhea (to support that there are "losses").

93:

In patients with infective endocarditis, the primary goal of anti-infective therapy is to effectively eradicate the pathogen. The duration of anti-infective treatment will vary based on clinical factors such as the pathogen, the type of valve, vegetation burden including area and penetration of infection, and if the patient has a history of intravenous substance use disorder. The duration of anti-infective treatment can last between 2 to more than 6 weeks, depending on the clinical factors mentioned above. In unique scenarios where the bactericidal activity against the pathogen is rapid, the vegetation burden is smaller, and the affected valve is a native valve, a shorter duration of anti-infectives could be prescribed (Delgado et al., European Heart Journal 2023: epub; Baddour et al., Circulation 2015, 132: 1435-86). However, certain patients, such as those with history of intravenous substance use disorder, may need to be treated longer due to the increased risk of reinfection, those with prosthetic valve endocarditis, persistent fever, history of endocarditis within the past year, immunocompromised, or resistant organism(s). The first day of a confirmed negative blood culture is important to document as it influences the total duration of anti-infectives and is considered the start of the recommended anti-infective duration countdown. In all cases, an Infectious Disease consultation as part of the multidisciplinary team is highly recommended to assist in determining the most appropriate anti-infective regimen (Sandoe et al., Heart 2023: epub; Luque Paz et al., Expert Rev Anti Infect Ther 2021, 19: 297-307; Otto et al., Circulation 2021, 143: e72-e227).

94:

Recommendations regarding evaluation, treatment, and setting of care for patients with fever and neutropenia are based on risk stratification into low and high risk groups. Assessment of the risk for complications from a severe infection should be performed upon presentation.

Low risk patients are generally defined by clinical stability without an active medical comorbidity (e.g., mucositis, nausea and vomiting, mental status change, etc.). High risk factors may include, but are not limited to, the following:

- Absolute neutrophil count (ANC) $\leq 100/\text{cu.mm}(100 \times 10^6/\text{L})$
- Mental status change
- Chronic lung disease

- Abdominal pain, new onset
- Mucositis
- Nausea and vomiting
- Diarrhea
- Uncontrolled pain
- Underlying malignancy with bone marrow involvement (e.g., acute leukemia)

In addition, use of validated risk assessment tools which take into account a multitude of risk factors, such as the Multinational Association for Supportive Care in Cancer (MASCC) scoring system, are recommended to categorize patients into low and high risk groups. Pediatric guidelines specifically recommend that hospitals adopt and use a validated risk stratification tool to guide routine management. Select low risk patients may be candidates for outpatient management, but inpatient management remains the standard of care (Pettit et al., Am J Emerg Med 2021, 50: 5-9; Zimmer and Freifeld, J Oncol Pract 2019, 15: 19-24).

95:

An absolute neutrophil count that is expected decline to $< 500/\text{cu.mm}(500 \times 10^6/\text{L})$ within 48 hours is included in the definition of neutropenia (National Comprehensive Cancer Network, The NCCN clinical practice guidelines in oncology, Prevention and Treatment of Cancer-Related Infections (Version 2.2023). 2023).

96:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

97:

Instruction: This criterion refers to a long-term, continuous electroencephalogram (EEG) which may last less than or equal to 72 hours and may or may not include video.

98:

Preeclampsia with severe features is classified by the American College of Obstetricians and Gynecologists (ACOG) as one of the following (Obstet Gynecol 2020, 135: e237-e60):

- Systolic blood pressure (BP) of at least 160 mmHg and/or diastolic BP of at least 110 mmHg on 2 or more measurements at least 4 hours apart (unless antihypertensive therapy is initiated prior to this time, severe hypertension can be confirmed within minutes to facilitate timely intervention)
- Systolic BP of at least 140 mmHg and/or diastolic BP of at least 90 mmHg on 2 or more measurements at least 4 hours apart and one of the following:
 - New-onset headache not related to alternative diagnoses and unresponsive to medication
 - Visual disturbances
 - Pulmonary edema
 - Liver dysfunction indicated by abnormally elevated liver enzymes to more than twice the upper limit of normal or by severe persistent right upper quadrant or epigastric pain unresponsive to medication
 - Renal insufficiency as evidenced by a serum creatinine greater than 1.1 mg/dL or serum creatinine that is doubled in the absence of other renal diseases
 - Thrombocytopenia with platelet count less than $100 \times 10^9/\text{L}$

InterQual® external peer reviewers recommend a higher level of care for individuals who present with pulmonary edema.

99:

Postpartum endometritis, an infection of the uterus after delivery, is diagnosed clinically based on the presence of fever and other infection-related symptoms in the absence of other causes (Izewski et al., Obstet Gynecol Clin North Am 2023, 50: 137-50). Cesarean delivery is the most significant risk factor for postpartum maternal infection. Individuals who undergo cesarean delivery have a 5- to 20-fold higher risk of infection and infectious morbidity

than those who deliver vaginally. Endometritis is one of the most common infectious complications following cesarean delivery (Caissutti et al., *Obstet Gynecol* 2017, 130: 527-38). Individuals with postpartum endometritis may exhibit fever, malodorous lochia, and pelvic discomfort. During the examination, individuals may present with uterine tenderness and display variable degrees of vaginal discharge. It is also important to note that patients may present with endometritis without substantial discharge (Wheaton et al., *Emerg Med Clin North Am* 2019, 37: 277-86). Individuals with endometritis may also experience abrupt or heavy vaginal bleeding or persistent or increased vaginal bleeding due to subinvolution of the uterus (National Institute for Health and Care Excellence: Clinical Guidelines, Postnatal care. 2021; Committee on Practice, *Obstet Gynecol* 2017, 130: e168-e86). Patients who have undergone a cesarean section may have evident external wound infections. Alternatively, they may also have a normal external examination without external wound infections. Administration of antibiotics and monitoring for and averting complications are the main goals of endometritis management. In certain circumstances, individuals with mild illness may be discharged home with oral antibiotics and close monitoring; however, the most prudent course of action is admission for intravenous antibiotics. Upon initiation of antibiotic treatment, it has been observed that roughly 90%(0.90) of individuals exhibit a reduction in fever within 48 to 72 hours (Wheaton et al., *Emerg Med Clin North Am* 2019, 37: 277-86). It is suggested that antibiotics should be continued for at least 24 to 48 hours after complete resolution of clinical signs and symptoms (Izewski et al., *Obstet Gynecol Clin North Am* 2023, 50: 137-50).

100:

Postpartum hypertension (HTN) and preeclampsia refer to the presence of either persistent or exacerbated HTN in individuals who have previously experienced hypertensive disorders of pregnancy or a new-onset development of the condition (*Obstet Gynecol* 2020, 135: e237-e60). The occurrence of elevated blood pressure during the postpartum period is frequently observed in individuals who had hypertensive disorders of pregnancy during the antepartum period. However, it is also possible for high blood pressure to emerge for the first time following childbirth. Postpartum elevations in blood pressure represent a potential threat to maternal health (Hauspurg and Jeyabalan, *Am J Obstet Gynecol* 2022, 226: S1211-S21). HTN has emerged as a prominent factor contributing to postpartum readmission in the United States. Furthermore, it is worth noting that more than one-half of maternal deaths occur during the postpartum period. In addition to increased blood pressure, individuals with postpartum preeclampsia frequently present with new-onset headaches or visual disturbances. Timely recognition and intervention are crucial since these individuals are highly susceptible to many medical complications, including stroke, seizures, pulmonary edema, renal failure, congestive heart failure, and mortality (Khedagi and Bello, *Cardiol Clin* 2021, 39: 77-90). Half of all intracerebral hemorrhage cases attributed to preeclampsia manifest during the postpartum period. A significant number of difficulties related to HTN and preeclampsia manifest during the postpartum period (e.g., 44%(0.44) of eclamptic seizures and 10%(0.10) of maternal deaths) (Asali et al., *Pregnancy Hypertens* 2019, 17: 133-7).

The existing body of medical literature primarily concentrates on antenatal and intrapartum treatment, with limited available data pertaining to the postpartum period (Hauspurg and Jeyabalan, *Am J Obstet Gynecol* 2022, 226: S1211-S21; Khedagi and Bello, *Cardiol Clin* 2021, 39: 77-90; Suresh et al., *Obstet Gynecol* 2021, 138: 777-87; Asali et al., *Pregnancy Hypertens* 2019, 17: 133-7). During the initial postpartum period, there are noticeable changes in fluid distribution within the body, which are closely linked to variations in blood pressure. During the first 48 hours following childbirth, there is an initial decline in intravascular fluid levels, followed by a subsequent rise between postpartum days 3 and 6 when the mobilization of fluids occurs. Contributing factors include the administration of intravenous fluids and the loss of pregnancy-associated vasodilation (Khedagi and Bello, *Cardiol Clin* 2021, 39: 77-90). Several drugs and substances often utilized during the postpartum period have the potential to exacerbate HTN through three primary mechanisms: retention of fluids, activation of sympathomimetic responses, and direct constriction of blood vessels. Of particular significance are nonsteroidal anti-inflammatory medicines (NSAIDs), which are commonly recommended as analgesics in the postpartum period. These pharmaceutical agents reduce the production of prostaglandins, resulting in a diminished capacity for vasodilation and an augmented tendency for sodium retention. It is recommended that NSAIDs be prioritized over opioid analgesics. However, it is essential to consider that individuals with chronic HTN may need increased blood pressure monitoring and modifications to their treatment regimen when using these medications (*Obstet Gynecol* 2020, 135: e237-e60).

101:

Initiation of antihypertensive treatment as soon as reasonably possible is recommended for acute-onset severe hypertension (i.e., systolic blood pressure (BP) of 160 mmHg or more or diastolic BP of 110 mmHg or more, or both) that is persistent for 15 minutes or longer (Louis et al., *American Journal of Obstetrics and Gynecology* 2022, 227: B2-B24; *Obstet Gynecol* 2020, 135: e237-e60; *Obstet Gynecol* 2019, 133: e26-e50; Committee on Obstetric, *Obstet Gynecol*

2017, 129: e90-e5). Drug selection is individualized based on contraindications and risk for adverse outcomes. When BP is under control, oral maintenance medication is initiated to anticipate further treatment needs and prevent a recurrence of severe acute hypertension (Obstet Gynecol 2019, 133: e26-e50). If first-line agents fail to treat severe hypertension effectively, consultation with maternal-fetal medicine, internal medicine, anesthesia, or critical care subspecialists is recommended to discuss second-line agents (Obstet Gynecol 2019, 133: e26-e50; Committee on Obstetric, Obstet Gynecol 2017, 129: e90-e5).

102:

Magnesium sulfate has been shown to reduce the frequency and severity of seizures in pregnant individuals with severe preeclampsia. The rate of seizures in pregnant individuals with severe features of gestational hypertension or preeclampsia is 4 times higher than in those without severe features (e.g., 4 per 200 versus 1 per 200); thus, magnesium sulfate is recommended for the prophylaxis and prevention of seizures in this population (Obstet Gynecol 2020, 135: e237-e60).

103:

Chest tubes are used to evacuate air, fluid, or blood from the pleural space. For malignancies involving the pleura, chest tubes may also be used for the administration of sclerosing agents. Serial chest x-rays may be performed to ensure that there is no re-accumulation of air or fluid. The chest tube may be removed when there is no evidence of air leak, output has slowed to 250-400 mL/day, and the lung is fully expanded (Vuong et al., Respir Med 2018, 137: 152-66).

104:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus. VBG should not be used to determine oxygenation. Patients with chronic CO_2 retention hospitalized for acute respiratory illness will exhibit increased PCO_2 levels. PCO_2 values should be compared with baseline values when available and other signs and symptoms of respiratory distress should be considered when determining the appropriate level of care (e.g., worsening dyspnea, high respiratory rate, decreased oxygen saturation, confusion, drowsiness) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; McKeever et al., Thorax 2016, 71: 210-5).

105:

Patients admitted with pulmonary exacerbations may not return to prior respiratory function with treatment. At risk patients for permanent pulmonary decline include those with advanced disease and multiple prior exacerbation episodes (Cogen et al., Pediatrics 2017, 139:).

106:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2 . The following are estimates of O_2 delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

107:

Not all patients will recover full pulmonary function with treatment prior to discharge. A plateau is reached after the active phase of illness has passed and there is no significant improvement or change within a 2 day time frame despite treatment and weaning efforts.

108:

Instruction: For this criteria point, repositioning refers to adjustment of the chest tube.

109:

Chronic obstructive pulmonary disease (COPD) exacerbations should be treated with an anti-infective in the following circumstances:

- When there is increased dyspnea, sputum volume, and sputum purulence
- When there is increased sputum purulence and either increased dyspnea or sputum volume
- When mechanical ventilation is required

(Rodriguez-Roison, 2013; i-xvi; 1-80).

Anti-infective therapy, when administered within the first 2 days of COPD exacerbation, is associated with lower morbidity, a reduced incidence of treatment failure, a decreased duration of mechanical ventilation, and a lower rate of readmission (Rothberg et al., JAMA 2010; 303(20): 2035-2042).

110:

Many patients with presumed chronic obstructive pulmonary disease (COPD), as a cause of dyspnea, do not respond to treatment for COPD. Those who are slow to respond to initial therapy may benefit from an evaluation by a clinical specialist such as a pulmonologist and should be re-evaluated for other medical conditions including pneumonia, heart failure, pneumothorax, pleural effusion, pulmonary embolism, and cardiac arrhythmia. Additional testing may include (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

- Chest x-ray can be used to indicate a diagnosis of pneumonia, pleural effusion, heart failure, or pneumothorax
- Electrocardiogram (EKG) can identify patients with right heart hypertrophy, arrhythmias, and ischemic episodes
- A spiral computed tomography (CT) scan may assist in identifying patients with pulmonary embolus
- B-type natriuretic peptide (BNP) may help to differentiate between heart failure and COPD

111:

Instruction: These criteria apply to the administration of a bronchodilator via a nebulizer, an inhaler, or an intravenous route. According to the available evidence, inhalers produce outcomes that are at least equivalent to nebulizers. Further research is needed to determine the optimal inhalation device according to individual patient characteristics and disease state. Education on correct usage is vitally important to ensure effective medication

delivery to the lungs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; van Geffen et al., Cochrane Database Syst Rev 2016: CD011826).

112:

The administration of systemic corticosteroids in patients with a chronic obstructive pulmonary disease (COPD) exacerbation improves lung function, decreases hypoxemia, shortens recovery time, reduces hospital length of stay, and prevents early reoccurrence. Evidence demonstrates daily administration of 40 milligrams of prednisone in patients who have been hospitalized for more than 5 days is not inferior to a 14-day course in terms of early relapse. This shortened regimen is supported by current guidelines. Studies have shown low-dose oral steroids demonstrate equivalent outcomes to intravenous therapy in the treatment of COPD. Nebulized budesonide may be an alternative treatment for exacerbations with similar benefits to intravenous methylprednisolone in select patients, according to the GOLD guidelines (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

113:

The goal of oxygen titration in the setting of a chronic obstructive pulmonary disease (COPD) exacerbation is to maintain baseline oxygen saturation, provide adequate oxygenation, and prevent worsening hypercarbia. Guidelines recommend that oxygen should be titrated to achieve an oxygen saturation between 88%(0.88) and 92% (0.92) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

114:

Oximetry (SpO₂) is used to monitor oxygenation status and guide initiation or adjustment of supplemental oxygen therapy. Arterial blood gas (ABG) analysis should be performed in the following circumstances (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

- Oxygen saturation below 92%(0.92)
- Suspected respiratory failure
- Prior to initiation of invasive or non-invasive mechanical ventilation
- 30 to 60 minutes after the initiation of supplemental oxygen therapy, and serially thereafter
- Prior to hospital discharge in patients with hypoxemia during the exacerbation

115:

Airway clearance therapy (ACT) is used to optimize pulmonary function by clearing sputum from the respiratory tract. The most effective techniques are:

- Chest physiotherapy (CPT): Percussion and vibration of the chest and can be combined with modified postural drainage (body positioning without the head down) to remove mucus from the lungs.
- Positive expiratory pressure (PEP): A mask or mouthpiece and breathing against applied pressure (10-20 cm H₂O). Oscillating PEP (not recommended in children), vibrates the airflow during PEP
- Active-cycle-of-breathing technique (ABCT): A cycle of deep controlled breathing, and forced thoracic expiration to clear secretions (used in all stages of disease)
- Autogenic drainage (AD): A method of controlled breathing to move mucus from the small airways to the larger airways for expulsion
- High-frequency chest compression (use of a compressive vest to vibrate the airways)

There is insufficient evidence to recommend one technique over another, and treatment should be individualized according to age, patient preference, and risk for adverse effects. The Cystic Fibrosis Foundation recommends that airway clearance therapy increase in duration or frequency during the treatment of pulmonary exacerbations (Strickland et al., Respir Care 2015, 60: 1071-7).

116:

Contraindicated refers to **documented** clinical circumstances when a medication or treatment is not indicated based on the nature of the disease, comorbid condition, or a drug allergy or intolerance.

117:

After a patient has been weaned from ventilatory support (e.g., mechanical or noninvasive positive pressure ventilation (NIPPV)), an additional day may be necessary to monitor for respiratory complications before transfer to a lower level of care.

118:

Although the vast majority of wound care is readily handled in the post-acute setting, there are situations when a patient requires care at the acute level, such as high-frequency wound care required, prolonged length of time required performing the procedure, and the type or amount of medication required to keep the patient comfortable during the procedure. For children less than 12 years of age (or less than 30 kg), the wound care regimen may need to be adjusted to a lesser frequency and/or duration (e.g., twice per day, 10 minutes), though still requiring the current level of care, due to the patient's size, pain tolerance, fluid shifts, and vital sign lability.

To determine the appropriate discharge destination, an assessment is required of the patient and/or caregiver's ability to learn the necessary wound care regimen, complexity of the treatment regimen, and a plan for managing contributory factors. For discharge to the home setting, the patient and/or caregiver must be able and willing to participate in the care plan, or there is sufficient agency assistance available to adequately meet the patient's needs. Patient and/or caregiver instruction should include all the following (Bryant and Nix, *Acute & chronic wounds: current management concepts*, 5th ed. 2015, xiv, 655 p.):

- Wound care regimen that includes a plan to monitor patient-specific contributory factors
- Adjunctive therapies (medications, nutritional plan, and supplements)
- Re-ordering medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Continued follow-up and support with health care providers

119:

There have been significant improvements in immunosuppressant therapy for transplant recipients. Early protocols included azathioprine and corticosteroids, but with the addition of calcineurin inhibitors in solid organ transplantation, long-term survival and recovery has been improved. A multi-drug approach to therapy allows lower dosing of potentially very toxic individual agents. The 3 phases of immunosuppressive therapy include (Enderby and Keller, *The American journal of managed care* 2015, 21: s12-23):

- *Induction* - Immediately after transplant, when the risk for rejection is highest, intense antibody therapy or higher doses of maintenance therapy immunosuppressive agents are given depending on the organ being transplanted and the transplant center. Antibody therapies are T-cell depleting polyclonal (e.g., antithymocyte globulin (ATG)) and monoclonal (e.g., alemtuzumab or anti- CD52 antibody) or nondepleting agents (e.g., basiliximab).
- *Maintenance* - Transplant recipients remain on one or a combination of maintenance immunosuppressive therapy for the rest of their lives. Calcineurin inhibitors (e.g., cyclosporine, tacrolimus) are the mainstay of maintenance therapy with additional agents including antiproliferative/antimetabolites (e.g., mycophenolate mofetil (MMF), mycophenolic acid (MPA)), corticosteroids, mammalian target of rapamycin (mTOR) inhibitors (e.g., sirolimus), or T-cell costimulation blockers. The most common combination of these post transplant (kidney, liver, lung, heart, or pancreas) include tacrolimus, MMF/MPA, and prednisone. Immunosuppressant therapy may cause opportunistic infections, cancers and other adverse side effects. To limit these effects and increase survival and quality of life, research is underway that includes the infusion of marrow-derived autologous mesenchymal stem cells (MSCs), xenotransplantation, and advancements in regenerative medicine.
- *Rejection* - Anti-rejection therapy is given to reverse an episode of rejection and is based on the severity and type of rejection, as well as the hemodynamic consequences of rejection. Treatment includes immunosuppressant medications given for treating an acute rejection during the initial post-transplant and may consist of a medication adjustment, steroid administration, or the addition of T-cell depleting or nondepleting agents.

120:

Medication adjustments tend to be driven by patient response. Patients being treated for symptoms of rejection usually have a biopsy and their medications adjusted based on the biopsy results to control the rejection.

121:

All medications commonly used to prevent rejection or opportunistic infections in the post-transplant period cause some degree of side effects. One of the important tasks of the transplant team is to titrate the dosage to minimize toxicity while maximizing benefit. Often this fine-tuning is empiric. If a patient has an intolerable side effect from a particular drug, an alternative medication or different combination or dosage of medications is attempted. Tacrolimus and cyclosporine (CSA) are immunosuppressive agents that inhibit calcineurin and are given to solid organ transplant recipients to prevent and treat allograft rejection. Tacrolimus is less nephrotoxic and may be more cost-effective than CSA. However, new-onset diabetes should be closely monitored during the medication period. The dosing of CSA is more difficult to adjust appropriately in children, especially in younger/smaller children (less than 2 years). This is thought to be due to their shorter intestine, leading to less absorption of the drug, and also their faster metabolism. More time may be needed to adjust the dose and determine the appropriate medication management in this population than in older children or adults. The goal is to preserve the graft without seriously harming the patient.

122:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

123:

Patients discharged from Observation should receive outpatient evaluation with echocardiogram (ECG) and early follow-up with their cardiologist within 3 days of discharge (Atzema and Singh, Cardiol Clin 2018, 36: 141-59).

124:

Administration of angiotensin-converting enzyme inhibitors and angiotensin receptor blockers decreases the recurrence of atrial fibrillation (Han et al., J Cardiovasc Pharmacol 2013, 62: 405-15; Li et al., Int J Clin Pract 2013, 67: 536-43).

125:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

126:

Instruction: Examples of antiarrhythmics with proarrhythmic risk include, but are not limited to, amiodarone, disopyramide, dofetilide, dronedarone, flecainide, mexiletine, propafenone, and sotalol. Many factors influence the appropriate choice of drug, including underlying medical conditions and concurrent drug administration. In some instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or

have limited evidence to support their use.

127:

Instruction: Patients with end-stage liver disease are generally on chronic outpatient lactulose, rifaximin, or neomycin at home to control symptoms of hepatic encephalopathy (HE). Hospitalization may be appropriate to initiate or adjust the dosage of these medications due to worsening HE.

128:

Hepatorenal syndrome (HRS) has been defined as acute renal failure in the presence of cirrhosis that manifests in profound circulatory dysfunction. A recent update in the diagnosis of HRS focuses on changes in serum creatinine levels instead of a fixed high serum creatinine value. Standard treatment for HRS includes drugs that increase blood pressure along with the albumin, a protein that is a volume expander. Per HRS guidelines, terlipressin with albumin is the recommended treatment, although in the U.S., because the use of terlipressin has not yet been approved, octreotide and midodrine are used in combination with albumin. A Cochrane Database Systematic Review analyzing the benefit or harmful effects of terlipressin given in combination with albumin found insufficient evidence in several small randomized controlled trials to support or refute terlipressin's superiority. Terlipressin was also associated with an increased risk of abdominal pain or diarrhea (Israelsen et al., Cochrane Database Syst Rev 2017, 9: Cd011532). Norepinephrine (also named noradrenaline) with albumin has been studied in relation to terlipressin and albumin and found to have a comparable efficacy. The use of norepinephrine in adult patients was validated by InterQual® consultants. Both regimens showed about a 40-50% response rate and up to a 50% rate of relapsing post-treatment. There is a high morbidity and mortality associated with HRS. Liver transplant screening is recommended for patients with this syndrome (Piano et al., Hepatol Int 2018, 12: 122-34; Acevedo and Cramp, World journal of hepatology 2017, 9: 293).

129:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2 . The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

130:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

131:

Hyperkalemia may cause electrocardiogram (ECG) changes such as a tall, narrow, T wave and a widening of the QRS complex (Sandau et al., Circulation 2017, 136: e273-e344).

132:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

133:

Weaning from mechanical ventilation is the transition from ventilator support to spontaneous, unassisted breathing. Spontaneous breathing trials (SBT) are the preferred method of weaning, although, the best mode and timeframe are disputed.

Two types of SBT are:

- 30 minutes of pressure support ventilation (PSV)
- 2h of T-piece ventilation

Weaning as soon as possible may shorten the patient's length of stay and minimize the risk of complications (e.g., ventilator-associated pneumonia, deep venous thromboembolism, deconditioning) (Subira et al., *Jama* 2019, 321: 2175-82; Blackwood et al., *Cochrane Database Syst Rev* 2014, 11: CD006904).

Weaning interventions may include:

- Reduction in ventilator support
- Reduction in FiO₂
- Increased length of SBT
- Increased number of SBT

134:

Pericardial diseases that are prone to developing cardiac tamponade include (Mekontso Dessap and Chew, *Intensive Care Med* 2018, 44: 936-9):

- Human immunodeficiency virus infection
- Tuberculosis
- Malignancies
- Renal failure
- Trauma/iatrogenic
- Hemopericardium in aortic dissection
- Rupture of the heart after acute myocardial infarction
- Bacterial infections
- Autoimmune disorders
- Radiation-induced
- Pneumopericardium
- Large (> 2 cm) idiopathic chronic pericardial effusions (which occur mostly in individuals whose sex at birth is female and who are at least 50 years of age)

135:

Patients that are at high risk for cardiac tamponade and not exhibiting symptoms can be monitored at the Intermediate level of care. These patients are hemodynamically stable; however, the pericardial effusion size is moderate (1cm to less than 2cm) to large (greater than 2 cm) requiring more intensive monitoring and potentially the need for serial echocardiograms (Mekontso Dessap and Chew, *Intensive Care Med* 2018, 44: 936-9; Adler et al., *Eur Heart J* 2015, 36: 2921-64).

136:

The medical therapy used to treat pericarditis/pericardial effusions depends on the etiology of the effusion. Medical therapy for these patients often involves treatment with NSAIDs, corticosteroids, and colchicine (Adler et al., *Eur Heart J* 2015, 36: 2921-64).

137:

Transthoracic echocardiogram is the initial modality used in the evaluation of suspected pericardial disease. It is inexpensive, rapidly performed, and available in virtually all hospital settings (Doherty et al., *J Am Coll Cardiol* 2019, 73: 488-516; Expert Panel on Cardiac et al., *J Am Coll Radiol* 2017, 14: S127-S37; Adler et al., *Eur Heart J* 2015, 36: 2921-64; Cosyns et al., *Eur Heart J Cardiovasc Imaging* 2015, 16: 12-31; Klein et al., *J Am Soc Echocardiogr* 2013, 26: 965-1012 e15).

138:

Cardiac transplantation provides the greatest survival benefit and remains the gold standard treatment in patients

with heart failure (HF), but limited availability of donor hearts make ventricular assist devices (VAD) a viable alternative. A VAD is a mechanical pump that assumes the function of the damaged ventricle and helps restore normal hemodynamics and end-organ blood flow. Pumps provide either pulsatile or continuous flow and may be implanted in one or both ventricles:

- Bi-ventricular assist device (BiVAD)-replaces the function of both ventricles
- Left ventricular assist device (LVAD)-replaces the function of the left ventricle
- Right ventricular assist device (RVAD)-replaces the function of the right ventricle

VAD insertion is currently approved as:

- Bridge to recovery: in patients with reversible ventricular dysfunction after cardiac surgery or myocardial infarction (MI)
- Bridge to transplantation: as temporary support for patients with end-stage HF awaiting transplant
- Destination therapy: as a permanent cardiac replacement device for patients who are not candidates for heart transplant

There are some devices that are not portable and require the patient to remain hospitalized while the device is in place. Some VADs can be inserted percutaneously and therefore, may be used more quickly than traditional, more invasive VADs (Miller and Rogers, JAMA Cardiol 2018;; Mancini and Colombo, Journal of the American College of Cardiology 2015, 65: 2542-55; Maini et al., Catheter Cardiovasc Interv 2014, 83: E183-92).

139:

In some facilities a ventricular assist device (VAD) will be used as a bridge to heart or heart-lung transplant. This may stabilize a patient long enough to survive until transplant, but the patient is still fragile and relatively unstable.

Instruction: In this clinical setting, the patient may require Critical care until a donor organ is available. Due to the restrictions associated with these devices, the Critical level of care is required for percutaneous VADs.

140:

There are 2 types of thrombi which can develop in a ventricular assist device (VAD) (Bartoli et al., J Card Surg 2014, 29: 83-94):

- A red thrombus occurs when stagnating blood coagulates due to low pressure and is formed when fibrin deposits trap erythrocytes. This type of thrombus is best managed with anticoagulation medications.
- A white thrombus occurs in areas of turbulence and is composed of activated platelets and is treated with antiplatelet agents.

141:

Absolute contraindications to thrombolytic therapy include prior or suspected intracranial bleed, prior stroke, intracranial aneurysm, arteriovenous malformation, neoplasm, closed head or facial trauma, active internal bleeding, suspected aortic dissection, known bleeding diathesis, or uncontrolled hypertension (greater than 180 systolic, greater than 100 diastolic) (Bartoli et al., J Card Surg 2014, 29: 83-94).

142:

Reference: (Rutledge et al., J Heart Lung Transplant 2013, 32: 569-73)

143:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

144:

Apheresis is a procedure that separates and removes elements in the blood (e.g., white blood cells (WBCs), platelets, stem cells, plasma bound organic compounds, or antibodies). Subsequent replacement with blood components, solvent detergent plasma (SDP), albumin, and/or normal saline may follow apheresis.

145:

Acute humoral rejection (AHR) is caused by antibodies that form within 2 to 7 days after transplant and presents with an acute loss of graft function. AHR is diagnosed by a C4D stain positive for circulating alloantibodies. Treatment is a combination of plasmapheresis (to remove antibodies) and heightened immunosuppression (tacrolimus, mycophenolate, steroids with muromonab-CD3 added) (Colvin, J Am Soc Nephrol 2007; 18(4): 1046-1056).

146:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

147:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children:

Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

148:

Refractory seizures, also known as uncontrolled, intractable, or drug-resistant seizures, are resistant to treatment and are poorly controlled with medication. Recurrent seizures are seizures that occur more than once and may occur without warning (Billington et al., J Clin Med 2016, 5:).

149:

Refractory arrhythmias, also known as uncontrolled, intractable, or drug-resistant arrhythmias, are resistant to treatment and are poorly controlled with medication. Recurrent arrhythmias are arrhythmias that occur more than once and may occur without warning.

150:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

151:

Blood pressure is considered labile when it is extremely variable and difficult to control.

152:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., *Kidney Int* 2020, 98: 294-309; Depret et al., *Ann Intensive Care* 2019, 9: 32; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii-138).

153:

Hyperosmolar therapy creates an osmotic gradient, facilitating the removal of water across the blood brain barrier, resulting in a decreased brain volume and a decrease in intracranial pressure. While hypertonic saline and mannitol are the 2 most used hyperosmolar agents, there is insufficient evidence to recommend one over the other for improving neurological outcomes in patients with acute ischemic stroke (Cook et al., *Neurocrit Care* 2020, 32: 647-66). Doses and strengths are generally determined by institutional practice patterns. Common side effects associated with osmotic therapy include dehydration, hypotension, renal failure, and acute respiratory distress.

154:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., *J Thromb Haemost* 2013:).

155:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., *American College of Cardiology Perspectives and Analysis* 2018; Napp et al., *Herz* 2017, 42: 27-44; Guirand et al., *J Trauma Acute Care Surg* 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., *J Am Coll Cardiol* 2014, 63: 2769-78; Domico et al., *Pediatr Crit Care Med* 2012, 13: 16-21; MacLaren et al., *Intensive Care Med* 2012, 38: 210-20; Peek et al., *Health Technol Assess* 2010, 14: 1-46; Mugford et al., *Cochrane Database Syst Rev* 2008: CD001340).

156:

A *Clostridioides difficile* infection (CDI) is considered fulminant (severe) when associated with any one of the

following (Kelly et al., Am J Gastroenterol 2021, 116: 1124-47):

- Hypotension/shock
- Ileus
- Megacolon

Instruction: Patients who are diagnosed with fulminant CDI and are hypotensive/exhibiting symptoms of shock or have a confirmed megacolon will meet at the Critical level of care in this subset.

157:

Due to the lack of normal peristalsis in the intestine, an ileus can slow down the absorption of oral vancomycin. Vancomycin delivered via enema is an alternative route that may offer greater benefit of absorption (Kelly et al., Am J Gastroenterol 2021, 116: 1124-47; McDonald et al., Clin Infect Dis 2018).

158:

Within the first 72 hours, volume resuscitation and vancomycin (oral or rectal) are the initial treatments of fulminant *Clostridioides difficile* infection. In addition to vancomycin (oral or rectal), intravenous metronidazole should be added when an ileus is present (Kelly et al., Am J Gastroenterol 2021, 116: 1124-47).

159:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

160:

Reference: (Jordan and Caesar, BMJ Qual Improv Rep 2015, 4:)

161:

The "R on T phenomenon" is the superimposition of an ectopic beat on the T wave of a preceding beat. Early observations suggest that R on T is likely to initiate sustained ventricular tachyarrhythmias.

162:

When T and U waves fuse, they form a TU fusion wave on an electrocardiogram. Severe hypokalemia may be associated with a large U wave created by T and U wave fusion (Zieg et al., Acta Paediatr 2016, 105: 762-72; Levis, Perm J 2012, 16: 53).

163:

The U wave occasionally follows the T wave on an electrocardiogram. A prominent upright U wave is associated with hypokalemia. When the U wave exceeds the T wave amplitude, the serum potassium level is less than 3 mEq/L. Severe hypokalemia may be associated with a giant U wave created by T and U wave fusion (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72).

164:

Symptoms of hyponatremia may include nausea, lethargy, headache, and muscle weakness. If hyponatremia worsens, symptoms may progress to mental status changes (Glasgow Coma Scale (GCS) 9-14). Patients with acute coma, stupor, obtundation, and seizures should be reviewed at a higher level of care due to the need for more frequent neurological monitoring (Weismann et al., Endocr Connect 2016:).

165:

Symptoms of hyponatremia include:

- Coma
- Headache
- Lethargy
- Nausea

- Obtundation
- Seizures
- Stupor
- Vomiting

166:

Fluid restriction should be avoided during the first 24 hours of therapy with conivaptan or tolvaptan as it may increase the likelihood of an overly rapid correction of serum sodium and may lead to the development of cerebral edema.

167:

An intra-aortic balloon pump (IABP) is a mechanical circulatory support device that utilizes a catheter-mounted intravascular device positioned in the descending thoracic aorta. The balloon automatically inflates during early diastole, increasing coronary blood flow and deflates during early systole, reducing afterload as the left ventricle ejects. Indications for use include cardiogenic shock, temporary support of left ventricular function after myocardial infarction (MI), preservation of myocardial viability, or refractory ventricular arrhythmias post-MI or preoperative bridge before coronary artery bypass graft (CABG) surgery. Contraindications include aortic regurgitation and aortic dissection.

168:

Placement of an intra-aortic balloon pump (IABP) may be beneficial for patients with recurrent ischemia despite intensive medical management, mechanical complications of a myocardial infarction (MI), hemodynamic instability, or cardiogenic shock awaiting coronary angiography and/or revascularization (Amsterdam et al., *Circulation* 2014, 130: 2354-94; Anderson et al., *J Am Coll Cardiol* 2013, 61: e179-347; O'Gara et al., *Circulation* 2013, 127: e362-425; Thiele et al., *N Engl J Med* 2012, 367: 1287-96).

169:

When weaning is appropriate, patients should be evaluated at least daily for the potential to wean. There is no universal standard for ventilator weaning and practice can vary due to the patient's needs and practitioner strategy (Burns et al., *Jama* 2021, 325: 1173-84).

Ventilator modes during the weaning process may include:

- Intermittent mandatory ventilation (IMV)
- Synchronized intermittent mandatory ventilation (SIMV)
- Pressure support
- Continuous positive airway pressure (CPAP)
- T-piece with breathing spontaneously for increasing lengths of time
- Noninvasive positive pressure ventilation (NIPPV)
- Consensus guidelines recommend the use of NIPPV in patients at high risk for reintubation and who have passed an SBT (Girard et al., *Am J Respir Crit Care Med* 2017, 195: 120-33)

170:

Optimal timing for tracheostomy placement for a ventilated patient is controversial within the literature. Per the Lung-Safe study, patients that required prolonged ventilation had a median time of about 14 days to tracheostomy placement (Abe et al., *Crit Care* 2018, 22: 195). Although early tracheostomy may help prevent ventilator associated pneumonia, it was not associated with reduced ventilator time or overall mortality (Chorath et al., *JAMA Otolaryngol Head Neck Surg* 2021, 147: 450-9). Patients that are at risk for failing to wean from the ventilator are older, have at least 1 comorbid, and those with severe acute conditions (Torrini et al., *Crit Care* 2021, 25: 391).

171:

Failed weaning trial indicates failure or intolerance to the wean and the cause for this should be determined and corrected. Subsequent weaning trials should be performed every 24 hours once the reversible cause(s) for failure are corrected and the patient is medically stable.

172:

Cardiac tamponade is an emergent situation and transthoracic echocardiogram is recommended to evaluate the effusion and subsequent hemodynamic compromise (Doherty et al., J Am Coll Cardiol 2019, 73: 488-516; Adler et al., Eur Heart J 2015, 36: 2921-64; Cosyns et al., Eur Heart J Cardiovasc Imaging 2015, 16: 12-31).

173:

Management of pericardial effusions should be aimed towards treating the underlying disease (e.g., malignant, infectious, connective tissue disorder, tuberculosis, iatrogenic cause) whenever possible. Pericardiocentesis, or cardiac surgery, is indicated for cardiac tamponade or large symptomatic effusions unresponsive to medical therapy (Adler et al., Eur Heart J 2015, 36: 2921-64).

174:

Patients that are symptomatic (dyspneic, orthopneic, or have jugular venous distention) that present with a pericardial effusion may be in cardiac tamponade. Patients with these symptoms require Critical level of care for potential immediate intervention (Mekontso Dessap and Chew, Intensive Care Med 2018, 44: 936-9; Adler et al., Eur Heart J 2015, 36: 2921-64).

175:

Pericardial drainage and pericardiocentesis are the primary interventions for cardiac tamponade related to pericardial effusion or management of a large chronic pericardial effusion (Adler et al., Eur Heart J 2015, 36: 2921-64).

176:

Thrombolytic agents are medications that act to dissolve culprit clots. Indications for the use of thrombolytic agents include acute myocardial infarction, thrombotic stroke, massive pulmonary embolism, deep vein thrombosis, peripheral arterial thromboembolic occlusion, and vascular graft thrombosis. The most common complication of thrombolytic therapy is hemorrhage. Risk increases with the concomitant use of heparin or other antithrombotic agents. There is also a risk of embolism occurring from a dissolving clot.

177:

Toxic megacolon is defined as colonic dilation (greater than 6 cm) with three of the following symptoms:

- Fever
- Leukocytosis
- Tachycardia
- Anemia/bleeding

In addition to the above symptoms, the patient must have one of the following:

- Hypotension
- Hypovolemia
- Mental status change
- Electrolyte imbalance

Since hypokalemia and hypomagnesemia can exacerbate the dilation, these electrolyte imbalances should be aggressively corrected when present. Patients with toxic megacolon should be treated with intravenous hydration, broad-spectrum antibiotics, and bowel rest. Nasogastric decompression may be considered to remove air and fluid from the colon. Rapid deterioration, continued toxicity, or lack of improvement may be an indication for prompt colectomy. In most cases, surgery is performed after a short trial of medical therapy (Desai et al., Clin Exp Gastroenterol 2020, 13: 203-10; Gajendran et al., Disease-a-Month 2019:).

178:

A percutaneous left ventricular assist device (VAD) is used to protect the heart during invasive procedures or temporarily for short-term mechanical circulatory support while treatment decisions are made.

179:

Patients with a ventricular assist device (VAD) are at risk for developing hypovolemia, bleeding, infection, thromboembolism, and arrhythmias. Stability in patients with a VAD is often patient specific. Symptoms of instability in a patient with a VAD include fluctuating vital signs, decreased VAD flows, or decreased level of consciousness.

180:

Evidence of adequate perfusion includes intact mentation, adequate urine output, warm extremities, or the presence of peripheral pulses.

181:

In continuous flow axial pumps, measurement of systolic blood pressure is not possible because there is no audible closure of the aortic valve. The sound heard during auscultation represents the mean arterial pressure (MAP). Use of a Doppler flow probe is helpful in defining blood pressure in patients with a ventricular assist device (VAD). In patients with continuous flow pumps, the goal MAP is between 70 and 90 mmHg. The use of multiple vasoactive or antihypertensive agents may be required to achieve this goal (Mancini and Colombo, Journal of the American College of Cardiology 2015, 65: 2542-55; Wilson et al., J Am Coll Cardiol 2009, 54: 1647-59).

182:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

183:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

184:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

185:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)

- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

186:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

187:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

188:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

189:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

190:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed

professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

191:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

192:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

193:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

194:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the

InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

195:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

196:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

197:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

198:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

199:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

200:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

201:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

202:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784).

Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.